



Spatially-resolved cluster dynamics modeling of irradiation growth

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ABSTRACT

We develop here a spatially resolved, three-dimensional continuum model coupling cluster dynamics (SR-CD) and crystal plasticity to investigate irradiation growth in zirconium. The model uses scale separation to divide the population of the irradiation cluster into mobile and immobile families. Small interstitial and vacancy clusters are modeled using anisotropic reaction–diffusion equations. Among the immobile clusters, an atomistically-informed vacancy cluster to vacancy loop transition is taken into account. The coupling between the evolution equation of CD and the plastic deformation of the material is two-fold, with stress-informed bias factors and local inelastic strains computed from the evolution of the evolving cluster population. The numerical implementation of the model utilizes the finite element method to analyze both single-crystal and polycrystalline samples. The growth strains that are computed align well with the experimental data provided by Carpenter for single-crystal Zr. Furthermore, the transformation of a vacancy cluster into a complete vacancy loop, occurring at a size of 14 nm, is in agreement with experimental observations and atomistic simulations. The density, size, and growth rate of the dislocation loops, denoted as $\langle c \rangle$ and $\langle a \rangle$, also exhibit good agreement with transmission electron microscopy (TEM) analysis of irradiated Zr and its alloys. Our findings demonstrate that there is a spatial correlation between the growth of these dislocation loops and growth strains, significantly influenced by the crystal size. To explain the expansion of the $\langle a \rangle$ axis and the contraction of the $\langle c \rangle$ axis in irradiated Zr, it is necessary to consider the diffusion anisotropy difference (DAD) of mobile interstitial species. We show that the PWR Kearns parameters, specifically $f_r = 0.63$, $f_i = 0.32$, $f_a = 0.05$, confer enhanced irradiation resistance to Zr along the principal directions when compared to single crystals. Additionally, reducing the grain size to nanograins further enhances the resistance to irradiation-induced growth, particularly along the direction with the highest volume fraction of basal poles [0001].

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List of symbols and notations used in this paper.

Symbol	Description
J_M^k	Flux of mobile species k
c_M^k	Signed concentration of total individual defects contained in mobile cluster k
C_M^k	Unsigned concentration of total mobile clusters k
B_M	Matrix to convert mobile species concentrations from C_M^k to c_M^k
P_M^k	Production of mobile species k per lattice cite
c_I^k	Signed concentration of defects per lattice cite in immobile cluster k
C_I^k	Unsigned concentration of defects per lattice cite in immobile cluster k
B_I	Matrix to convert immobile species concentrations from C_I^k to c_I^k
P_I^k	Production of immobile species k per lattice site
$N^{\mathcal{L}}$	Number density of immobile species k
R_1	First order reaction matrix involving a single mobile species
R_2	Second order reaction matrix involving two mobile species
$\langle k_{\mathcal{L},m}^2 \rangle$	Sink strength of immobile species $\mathcal{L}$ to mobile species m
$\langle k_{d,m}^2 \rangle$	Sink strength of dislocation line d to mobile species m
$Z_{\mathcal{L},tot}^k$	Total bias factor of immobile species $\mathcal{L}$ to mobile species k
$Z_{\mathcal{L},k}^{DAD,0}$	DAD sink strength of immobile species $\mathcal{L}$ to mobile species k
Z_m^0	Absorption efficiency of an isotropic immobile species without stress
p^m	Diffusional anisotropy factor of mobile species m
$\overline{D}^m$	Average diffusion coefficient of mobile species m
$\alpha^{m,n}$	Dissociation rate of mobile species n into m
$\beta^{m,n}$	Interaction rate of mobile species n with m
$A^{\mathcal{L}}$	Average vectorial area of a dislocation loop $\mathcal{L}$
$b^{\mathcal{L}}$	Burgers vector of a dislocation loop $\mathcal{L}$
$r^{\mathcal{L}}$	Radius of a dislocation loop $\mathcal{L}$
T	Temperature of simulation domain
G_0	Dose rate
f_r	Fraction of recombining point defects from cascade
ε_k	Fraction of total interstitials produced as mobile species k
Ω	Atomic volume
$\Delta\Omega_k$	Relative relaxation volume of mobile species k
E_k^f	Formation energy of mobile species k
E_k^b	Binding energy of mobile species k
$E_k^{(ij)}$	Migration energies of mobile species k along direction ij
D_k	Diffusion coefficient of mobile species k
$g^{\mathcal{L}}$	Relative relaxation volume of immobile species $\mathcal{L}$
n_{min}	Critical number of vacancies below which immobile species is a bi-pyramid
n_{max}	Critical number of vacancies above which immobile species is a loop
$\Delta\Omega_{bp}$	Change in volume of a bi-pyramid defect for vacancy immobile species
α_{bp}	Bi-pyramid sink strength coefficient as a fraction of a void

1. Introduction

Irradiation growth is a phenomenon that describes permanent deviatoric dimensional changes in irradiated materials without externally applied stress (Was, 2016). It is typical in irradiated hexagonal closed-packed metals (HCP), especially zirconium (Zr) and its alloys, and it is characterized by expansion along the three $\langle a \rangle$ -axes and contraction along the $\langle c \rangle$ -axis of hexagonal crystals.

Under neutron irradiation, zirconium develops a highly complex and evolving defect microstructure, characterized by a mixture of mobile and immobile defects of both interstitial and vacancy types. The properties of self-diffusion and vacancy have been studied through experiments (Horvath et al., 1984; Hood, 1986, 1988), and immobile defects taking the form of dislocation loops in the basal and prismatic planes have been observed through transmission electron microscopy (TEM) in Zr irradiated with neutrons, ions, and electrons (Jostons et al., 1977; Northwood et al., 1979; Holt and Gilbert, 1983, 1986; Griffiths, 1988; Harte et al., 2017). Atomic simulations have been used extensively to study the accumulation of damage from displacement cascades in HCP Zr, as well as the properties of mobile and immobile defects (Mikhin et al., 1994; Gao et al., 2001; De Diego et al., 2002; Voskoboinikov et al., 2006; Peng et al., 2012; Varvenne et al., 2014; Ouyang et al., 2018; Christensen et al., 2018, 2020). The defect microstructure controls important macroscopic mechanical properties in Zr and its alloys, including irradiation growth and irradiation creep. While both phenomena result from the evolution of the defects microstructure, the key difference between them lies in their dependence on external stress: irradiation growth can occur without it, whereas irradiation creep specifically requires the application of stress to manifest. In the case of creep, the defects microstructure is known to be associated to a variety of deformation mechanisms, including radiation-retarded creep mechanisms, radiation-enhanced mechanisms, and radiation-induced mechanisms (Franklin et al., 1983).

Data on the growth of Zr and Zr alloys have recently been extensively summarized (Adamson et al., 2019). Experiments have shown that irradiation growth of annealed Zr single crystals consists of three distinct stages (Holt, 1988). Stage I shows a steep increase in strain from ~ 0.1 to ~ 1 displacements per atom (dpa). Stage II shows a very low strain rate, characterized as a strain plateau, and lasts up to ~ 3 dpa. Stage III, accompanied by an accelerated growth strain rate, occurs at a high dose and is termed the breakaway stage. These different growth strain rate behaviors are believed to be strongly correlated with the formation and growth of both prismatic $\langle a \rangle$ - and basal $\langle c \rangle$ -dislocation loops (Golubov et al., 2014). However, in cold-worked materials, strain rates are relatively high at the beginning of irradiation, and no strain plateau is formed (Zee et al., 1987; Choi and Kim, 2013). The influence of such changes on the microstructure is generally deleterious and is associated with a degradation in the longevity of the fuel cladding in light water reactors (LWRs), potentially compromising their safety.

Models of this phenomenon have been developed since the mid-sixties, with most of the proposed models based on the radiation damage rate theory (Bullough and Wood, 1980; Woo, 1988; Holt, 1988; Christien and Barbu, 2005, 2009; Choi et al., 2016; Li and Ghoniem, 2020; Zhang et al., 2023). The succession of rate theory models have progressively increased our understanding of the key roles of vacancy loops, growing on basal planes, and interstitial loops, growing on prismatic planes, in explaining the growth rate of single-crystal Zr. The growth from irradiation in annealed polycrystal Zr is approximately ten times greater than that of single crystal Zr (Rogerson, 1988; Adamson et al., 2019), and the modeling of the effects of grain boundaries as neutral and distributed sinks for point defects has been recognized as problematic. Existing irradiation growth models that incorporate grain boundary sinks do not include the effect of grain interaction stresses. Another important development in the theory of irradiation growth has been the concept of intrinsic interstitial anisotropic diffusion introduced by Woo and Goesele (1983). This concept allowed a description of irradiation growth in the presence of a mixed population of vacancy and interstitial $\langle a \rangle$ -type loops. Moreover, in the presence of collision cascades, the concept of production bias due to in-cascade clustering (Woo and Singh, 1992) was an important step to overcome shortcomings in attaining theoretically reasonable growth rates without the use of unrealistic dislocation bias values.

The current state of cluster dynamics modeling includes spatially resolved stochastic cluster dynamics (SR-SCD), and spatially resolved cluster dynamics (SR-CD). SR-SCD simulates individual events, such as the absorption of a mobile defect by a cluster, based on probabilities calculated from lower-scale models. The diffusion term associated with mobile defects is typically approximated to a reaction rate characterizing the migration of defects in neighboring volume elements. This term can then be incorporated into the stochastic cluster dynamics to model non-homogeneous microstructure and spatially correlated reactions between defects (Dunn et al., 2013, 2016). By explicitly considering the sink strength of grain boundaries, SR-SCD is also used to study the correlations between point and clustered defect behavior at material interfaces. However, due to the limitation of the computational efficiency where all types of defects need to be explicitly traced in every volume element, SR-SCD simulations are generally restricted to small simulation cells (~ 100 nm) and low dose regimes (< 0.1 dpa). On the other hand, SR-CD methods assume a continuous representation of defect clusters and their interactions through deterministic equations, leading to averaged-out behavior predictions that do not account for the discrete and random nature of defect interactions. The motivation to spatially resolve defect populations in cluster dynamics simulations stems from a variety of situations where interfaces play a significant role in the evolution of the defect microstructure. These include the defect clusters evolution in polycrystals, the behavior of thin films in *in-situ* irradiation experiments (Xu et al., 2012), and near-surface gas implantation in plasma-facing materials (Xu and Wirth, 2010; Maroudas et al., 2016). SR-CD models are typically based on finite difference or finite volume schemes, and have been limited to 1-D analysis (Faney and Wirth, 2014; Maroudas et al., 2016; Shah et al., 2020). A generalization of these models is challenging, mainly due to the limitation imposed by the grid size as well as the treatment of the full cluster population. Recently, Shah et al. (2020) succeeded in studying the helium kinetics in a thin film with a size up to ~ 100 μm resorting to reduced cluster dynamics equations and an adaptive grid method.

There is then a realization that the grain structure matters in a significant way when it comes to describing both irradiation growth and irradiation creep. Modeling grain boundaries as neutral as well as distributed point defect sinks has not only made it difficult to describe the irradiation growth and creep of polycrystal Zr, but also failed to provide a clear connection between growth and creep. Historically, irradiation creep and growth have been assumed to be separate and additive (Alexander et al., 1977; Holt and Ibrahim, 1979). The assumption is valid if the creep and growth mechanisms are totally independent. However, data have shown that the mechanisms of irradiation creep include the direct accumulation of strain from diffusional mass transport in addition to the dislocation glide mechanism (Adamson et al., 2019), and mechanistic steady-state creep models have been recently proposed (Xiao et al., 2022). A good description of all of thermal creep, in-reactor creep and in-reactor growth is thus necessary

to accurately describe strains in irradiated polycrystalline materials (Onimus et al., 2022). The ability to describe the direct role of grain structure, internal stresses, and the commingling of irradiation growth and creep poses a modeling challenge, and requires some form of coupling between radiation damage theory and the mechanics of inelastic deformation, which explicitly takes the grain structure into account.

The objectives of the present work are to develop a spatially resolved cluster dynamics (SR-CD) model for radiation damage in three dimensions (3D), to explicitly couple it to a mechanical model of inelastic deformation, to apply the model to investigate the phenomenon of irradiation growth in both single and polycrystal samples, and finally to verify the model using experimental data. To achieve these objectives, we set out to formulate a *weak form* of cluster dynamics (CD) equations suitable for finite element (FE) implementation in 3D, and suitable for direct coupling with inelastic deformation mechanics. The main advantage of this coupled model is that it naturally incorporates the important concept of anisotropic diffusion and can also be extended in the future to tackle many problems of materials in an irradiation environment where both stress and the neutron flux are present, for example, irradiation creep, irradiation hardening, and irradiation swelling. The model developed in this paper focuses on the investigation of the irradiation growth phenomenon observed in irradiated Zr and its alloys. Its possible extension to creep conditions requires incorporating creep-specific mechanisms, including the stress induced preferential absorption (SIPA) mechanism.

The organization of this paper is as follows. In Section 2, we develop a *reduced-order* SR-CD model of radiation damage. The model is first developed in Section 2.1, where the defect populations are divided into two groups, a mobile population and a sessile population. Then in Section 2.2, we describe the unique application of the model to zirconium irradiation growth, where defect anisotropic diffusion (DAD) is included (Woo, 1988), and discuss how we model the stages of irradiation growth. The mechanics of inelastic deformation is introduced in Section 2.3. The numerical implementation of the coupled model within the FE framework is then discussed in Section 3, where the weak form of the SR-CD equations is presented in Section 3.1, and the model parameters in Section 3.2. Results are presented in Section 4, where single crystals are analyzed in Section 4.1, the size effect in Section 4.2, the influence of diffusion anisotropy in Section 4.3, the behavior of polycrystals and the influence of texture in Section 4.4, as well as the expected growth behavior of nanograined polycrystals. Finally, the main conclusions of this work are presented in Section 5.

2. Model description

2.1. Spatially Resolved Cluster Dynamics (SR-CD) model

In irradiated materials, collision cascades produce a distribution of small interstitial and vacancy clusters. Depending on the material, a wide range of defect types including voids, bubbles, precipitates, and radiation induced loops are observed (Shi et al., 2022; Vo et al., 2023), however, in α -Zr the shape of defect clusters beyond a small size becomes planar, where they can be described as vacancy and interstitial dislocation loops (Onimus et al., 2020b). If gases such as hydrogen or helium are produced or if nuclear transmutations start to accumulate, the composition and shape of these defect clusters can become complex. When zirconium is irradiated in a fission reactor neutron spectrum, however, gases or other nuclear transmutations can be ignored (Adamson et al., 2019). The defect populations produced in irradiated Zr can be modeled using the so-called *mean field approximation*, where all spatial correlations between the clusters of defects are ignored. The corresponding species balance equations can be represented as a set of ordinary differential equations (ODEs) for defect concentrations, generically written as

$$\frac{dC^k}{dt} = \sum_{m \neq k, n \neq k} \left[\frac{1}{2} R_2(m+n \rightarrow k) C^m C^n - \frac{1}{2} R_2(m+k \rightarrow n) C^m C^k + R_1(m \rightarrow k) C^m - R_1(k \rightarrow m) C^k \right] + P^k \quad (1)$$

where C^k is the concentration of a generic defect species (i.e. point defect and defect clusters), with an index k that represents the number of interstitial atoms or vacancies contained in a cluster. The indices m, n are for all defect clusters containing m and n atoms, respectively. The first and second terms of Eq. (1) correspond to second-order reactions involving two reacting species m and n , leading to the removal and generation of species k , respectively. The prefactor $1/2$ is used to avoid double counting of the reactions. The third and fourth terms represent first-order reactions of conversion to and from species k , respectively. The R_2 term describes the interaction rates between different pairs of mobile species. The R_1 term encodes the reactions that involve a single mobile species, including self-dissociation and its annihilation at sinks such as immobile clusters, taking into account the effective diffusion rates and the sink strength of such sinks. Thus, from Eq. (1), $m+n \rightarrow k$ represents the interaction of mobile species m and n to form k , and $m \rightarrow k$ represents the absorption of mobile species m into immobile species k . P^k is the production rate of defect species k . It is therefore noted that a comprehensive dose-dependent description of defect evolution entails solving an exceedingly large number of coupled ODE systems, one equation for each cluster size.

To avoid the time-consuming computational efforts associated with a large number of ordinary differential equations (ODE) while preserving the precision of a model containing the evolution equations of all defects, a model with a reduced set of rate equations was proposed (Ghoniem, 1989), and recently extended to Zr (Li et al., 2020; Li and Ghoniem, 2020). In this reduced framework for defect kinetics, defects are categorized into two groups: small-sized, highly mobile clusters, which are individually solved, and large-sized, immobile clusters, which are approximated through various grouping methods describing their size distribution. In this work we also have included the diffusive flux term associated with mobile defects, which can account for their anisotropic diffusion, as well as a matrix description of the cluster dynamics equations, introducing a flexible framework to separate mobile and immobile species.

Mobile defects are characterized by an array containing the concentration of a mobile species $C_M = [\dots C_M^{2v}, C_M^v, C_M^i, C_M^{2i}, \dots]^T$, as well as the concentration of defects in a mobile species, $c_M = [\dots c_M^{2v}, c_M^v, c_M^i, c_M^{2i}, \dots]^T$ for divacancies (2v), single vacancy (v), single interstitials (i), di-interstitials (2i), and so forth. On the other hand, immobile species, such as dislocation loops, are characterized by their average radius $r_I^\mathcal{L}$, their number density $N_I^\mathcal{L}$, and the concentration of defects in an immobile species, $C_I^\mathcal{L}$ or $c_I^\mathcal{L}$, where $c_I^\mathcal{L}$ is signed to distinguish from vacancy and interstitial defects. For example $c_I^\mathcal{L} = [\dots c_{I1}^v, c_{I2}^i, c_{I3}^{2i}, \dots]^T$, corresponding to the total number of defects in immobile vacancy or interstitial clusters of type $I1, I2, I3$, etc. Mobile species concentrations can easily be converted to atomic defects concentrations by $c_M^k = B_M \cdot C_M$, where B_M is the diagonal matrix containing the number of vacancies or interstitials in each mobile cluster. The concentrations of immobile species follow a simple relation $c_I^\mathcal{L} = B_I \cdot C_I^\mathcal{L}$, where B_I represents the sign of the cluster (negative for vacancy type and positive for interstitial type). The conversion is explicitly shown as

$$\begin{bmatrix} \vdots \\ c_M^{2v} \\ c_M^v \\ c_M^i \\ c_M^{2i} \\ \vdots \end{bmatrix} = \begin{bmatrix} \ddots & & & & \\ & -2 & 0 & 0 & 0 \\ & 0 & -1 & 0 & 0 \\ & 0 & 0 & 1 & 0 \\ & 0 & 0 & 0 & 2 \\ & & & & \ddots \end{bmatrix} \cdot \begin{bmatrix} \vdots \\ C_M^{2v} \\ C_M^v \\ C_M^i \\ C_M^{2i} \\ \vdots \end{bmatrix} \quad \begin{bmatrix} \vdots \\ c_{I1}^v \\ c_{I2}^i \\ c_{I3}^{2i} \\ \vdots \end{bmatrix} = \begin{bmatrix} \ddots & & & & \\ & -1 & 0 & 0 & \\ & 0 & 1 & 0 & \\ & 0 & 0 & 1 & \\ & & & & \ddots \end{bmatrix} \cdot \begin{bmatrix} \vdots \\ C_{I1}^v \\ C_{I2}^i \\ C_{I3}^{2i} \\ \vdots \end{bmatrix} \quad (2)$$

We now extend the above formulation of Eq. (1) into spatially-resolved CD diffusion–reaction framework. Based on the above definitions, the governing equation for the atomic concentration of each *mobile* species, c_M^k , can be written as

$$\frac{\partial c_M^k}{\partial t} = -\nabla \cdot \mathbf{J}_M^k + \sum_{(m,n) \in M} \frac{B_M^k}{2} R_2^{k,mn} \frac{c_M^m}{B_M^m} \frac{c_M^n}{B_M^n} + \sum_{m \in M} \frac{c_M^n}{B_M^k} R_1^{k,m} c_M^m + B_M^k P_M^k \quad (3)$$

where B_M^k is the k th diagonal component in the matrix B_M , corresponding to the net interstitials in the mobile cluster. $\mathbf{J}_M^k = -\mathbf{D}^k \cdot \nabla c_M^k$ is the diffusion flux vector of a species k , $R_2^{k,mn}$ and $R_1^{k,m}$ characterizing the reaction rates of second order and first-order reactions, involving the production (or consumption) of the species k . The term P_M^k is the production bias of mobile species k generated directly from the displacement cascade.

Clusters larger than a critical size are treated as immobile sinks to mobile defects. The net interstitials [per lattice site] inside the *immobile* clusters, $c_I^\mathcal{L}$, can therefore be written as

$$\frac{\partial c_I^\mathcal{L}}{\partial t} = \frac{B_M^m + B_M^n}{2} R_2^{\mathcal{L}mn} \frac{c_M^m}{B_M^m} \frac{c_M^n}{B_M^n} + R_1^{\mathcal{L}m} c_M^m + P_I^\mathcal{L}, \quad (4)$$

where $B_M^m + B_M^n$ represents the net interstitials added to the immobile clusters, associated with the reaction of m and n mobile species. If we now apply the mean-field theory and assume that these clusters are very stable (no-dissociation), the above equation reduces to

$$\frac{\partial c_I^\mathcal{L}}{\partial t} = \frac{B_M^m + B_M^n}{2} R_2^{\mathcal{L}mn} \frac{c_M^m}{B_M^m} \frac{c_M^n}{B_M^n} + \langle k_{\mathcal{L},m}^2 \rangle \bar{D}^m c_M^m + P_I^\mathcal{L}, \quad (5)$$

where the first term represents the nucleation of large clusters by the reaction of mobile species ($m + n \rightarrow \mathcal{L}$). The second term is the absorption of mobile defects, characterized by a sink strength $\langle k_{\mathcal{L},m}^2 \rangle$ of the sink $\mathcal{L}$ to the mobile species m with an average diffusion coefficient $\bar{D}^m$. The last term in the above equation indicates the direct production from the radiation cascade; the so-called production bias (Woo and Singh, 1992).

From a geometric point of view, when dislocation loops are considered as the immobile sinks, the net interstitials (per lattice site) inside the loops $\mathcal{L}$ can be also expressed as

$$c_I^\mathcal{L} = (\mathbf{A}^\mathcal{L} \cdot \mathbf{b}^\mathcal{L}) N^\mathcal{L} = \pi (r^\mathcal{L})^2 b^\mathcal{L} N^\mathcal{L} \quad (6)$$

where $\mathbf{A}^\mathcal{L}$ is the average vectorial area of the loop and $\mathbf{b}^\mathcal{L}$ is the loop Burgers vector. The last equality in Eq. (6) is true only for pure prismatic loops, otherwise the angle between the loop normal and the burgers vector must be considered. Omitting the distribution of the loop radius and assuming the steady state with zero nucleation rate, i.e. $dN^\mathcal{L}/dt = 0$, the evolution of the average cluster size can be obtained by differentiating the above equation with respect to time and comparing to Eq. (5), which yields

$$\frac{dr^\mathcal{L}}{dt} = \frac{\langle k_{\mathcal{L},m}^2 \rangle \bar{D}^m c_M^m}{2\pi r^\mathcal{L} b^\mathcal{L} N^\mathcal{L}} \quad (7)$$

It is notable that Eq. (7) is consistent with the loop growth rate in the traditional rate theory formulation.

2.2. SR-CD of zirconium

We focus now on the irradiation growth of zirconium. Early irradiation growth experiments from Carpenter et al. (1981) observed an $\langle a \rangle$ axis expansion and $\langle c \rangle$ axis contraction of the hcp single crystal, that can be described by three distinct stages. This was

also observed in prior experiments by Buckley, and both experiments reported similar growth strains. The evolution of the defect microstructure is depicted in Fig. 1, showing the initial mobile defects on the left, and then the evolving microstructure during the three stages, progressing towards the right. During stage I, there is a rapid growth along the $\langle a \rangle$ direction and $\langle c \rangle$ directions. Differences in diffusion coefficients and migration energies along $\langle a \rangle$ and $\langle c \rangle$ directions for interstitials and vacancies leads to preferential formation of small interstitial clusters on prismatic planes. Vacancies diffuse slower than interstitials, and interstitial diffusion is enhanced in the basal plane from the diffusion anisotropy difference (DAD) (Woo, 1988), amplifying the formation of interstitial clusters in the basal plane (Christensen et al., 2020). Stage I is a nucleation stage and growth strain are small from the small clusters and loops, therefore we focus only on stage II and stage III. As irradiation increases to stage II, some SIA clusters grow large enough to form interstitial loops on the prismatic planes.

We do not consider vacancy prismatic loops during any growth stage as they cannot grow together with interstitial prismatic loops in rate theory (Barashev et al., 2015; Golubov et al., 2014). Recent TEM experiments have shown that at 623 K, only 10% of $\langle a \rangle$ loops are of vacancy type, and this percentage decreases with decreasing temperature (Liu et al., 2020). Furthermore, Object Kinetic Monte Carlo (OKMC) simulations showed that if $\langle a \rangle$ vacancy loops are capable of forming, then breakaway growth is not experienced, and there is a constant growth rate along both the $\langle a \rangle$ and $\langle c \rangle$ axes. However, if vacancy clusters are able to form into stacking fault pyramids, both a steady-state and breakaway growth stage are experienced (Sakaël et al., 2023). It has also been shown that at a damage rate of $1 \times 10^{-7} - 5 \times 10^{-7}$ dpa/s, few $\langle a \rangle$ vacancy loops form, regardless of the probability of vacancy clusters to transform into either $\langle a \rangle$ loops or stacking fault pyramids (Sakaël et al., 2023). Other OKMC models could also not reproduce experimental growth strains with the introduction of $\langle a \rangle$ vacancy loops (Christiaen et al., 2020). Even recent Molecular Dynamics (MD) simulations that have observed vacancy prismatic loops forming at stage II show that by stage III, they vanish (Christensen et al., 2020). For a simulation at a dose rate of 1×10^{-7} dpa/s and a temperature of 553 K used for this study, it is more than reasonable to assume there are no prismatic vacancy loops.

Along with the formation of the $\langle a \rangle$ prismatic interstitial loops, small immobile vacancy clusters begin to grow to form vacancy bi-pyramids at the beginning of stage II, as suggested by atomistic simulations and experiments (Christensen et al., 2020; Liu et al., 2020). The coexistence of the small vacancy bi-pyramids and prismatic loops keeps a nearly constant growth strain since the isotropic eigenstrain of the vacancy bi-pyramids compensates the expansion along the $\langle a \rangle$ direction. There is also a reduction of the net flux of the mobile species from the bi-pyramids. At the end of stage II, some of the bi-pyramids have grown large enough to transform into $\langle c \rangle$ vacancy loops on the basal planes, causing a net contraction along the $\langle c \rangle$ axis, as the eigenstrain of the defect has gone from isotropic to highly anisotropic. The increased growth strain rate as vacancy bi-pyramids collapse into loops is why stage III is termed the breakaway growth stage.

Based on the aforementioned growth mechanism, four mobile species and four types of immobile clusters (dislocation loops or volumetric defects) are considered in the SR-CD model developed here. The mobile species are: single vacancies (v), single interstitials (i), di-interstitials ($2i$), and tri-interstitials ($3i$). The immobile species are: three $\langle a \rangle$ -type interstitial loops ($a\mathcal{L}$), and either $\langle c \rangle$ -type vacancy loops ($c\mathcal{L}$) or vacancy bi-pyramids (bp). The rationale of this selection can be found in our previous papers (Li and Ghoniem, 2020; Li et al., 2020). Here, we briefly summarize the assumptions used in the present work:

1. Following a very fast transient (neutron damage dose ~ 0.1 dpa in Choi et al. (2016) and Li and Ghoniem (2020)), defects achieve a steady state with little variation in their concentrations. The nucleation of clusters larger than tri-interstitials can therefore be neglected after an initial “nucleation” dose, i.e. the number of large clusters has saturated.
2. Single vacancies, single interstitials, di- and tri-interstitials are directly generated from the radiation cascades. Single vacancies and single interstitials diffuse isotropically, while di- and tri-interstitials diffuse anisotropically.
3. SIA clusters with four or more interstitials are considered as $\langle a \rangle$ dislocation loops lying on the prismatic planes with burgers vector of a_1 , a_2 and a_3 . Vacancy clusters are either in the form of bi-pyramids or in $\langle c \rangle$ dislocation loops lying on basal planes, depending on their sizes. They can grow or shrink by absorbing mobile species m characterized by a sink strength $\langle k_{\mathcal{L},m}^2 \rangle$.
4. Based on recent atomic simulations (Christensen et al., 2020), small immobile vacancy clusters possess bi-pyramid type geometry, with a volume of $V_{bp} = \frac{\epsilon}{a} \sqrt{3} r_{bp}^3$, where r_{bp} is the radius of the cluster and c and a are lattice parameters. Upon growth of the pyramids, they collapse into planar loops on the basal plane with a volume of $V^{\mathcal{L}} = \pi (r^{\mathcal{L}})^2 b^{\mathcal{L}}$, if their size is greater than a critical value. This transition is characterized by a sigmoidal function $S(n_v)$, where n_v is number of vacancies per vacancy bi-pyramid. $S(n_v)$ approaches zero if bi-pyramids are dominant and vice versa for loops with $S(n_v) \sim 1$. Accordingly, the properties associated with vacancy clusters such as the sink strength and the radius are interpolated as $r_{cl}(n_v) = S(n_v)r^{\mathcal{L}}(n_v) + [1 - S(n_v)]r_{bp}(n_v)$, where $r_{cl}(n_v)$ is the radius of the interpolated cluster, $r^{\mathcal{L}}(n_v)$ is the radius of the dislocation loop and $r_{bp}(n_v)$ is the radius of the bi-pyramid.

Invoking quasi-static steady state for defect concentrations, cluster densities can be written as $C_M = [C^v, C^i, C^{2i}, C^{3i}]^T$ with $k = -1, 1, 2, 3$, and their steady-state mass conservation equations can be written in a compact array form,

$$0 = - \begin{bmatrix} \nabla \cdot \mathbf{J}_M^v \\ \nabla \cdot \mathbf{J}_M^i \\ \nabla \cdot \mathbf{J}_M^{2i} \\ \nabla \cdot \mathbf{J}_M^{3i} \end{bmatrix} + \frac{1}{2} \begin{bmatrix} C_M^T R_2^v C_M \\ C_M^T R_2^i C_M \\ C_M^T R_2^{2i} C_M \\ C_M^T R_2^{3i} C_M \end{bmatrix} + R_1 C_M + P_M \quad (8)$$

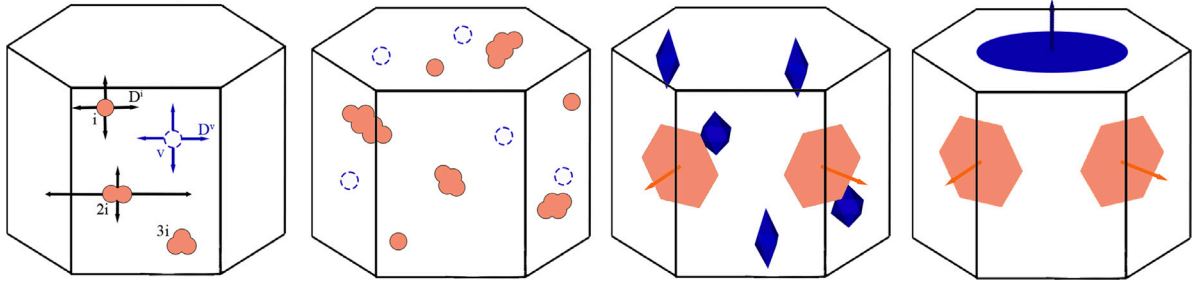


Fig. 1. Defect mobilities (left) and evolution of defect microstructure during three growth stages of single crystal α -zirconium.

where $\mathbf{J}_M^k = -\mathbf{D}^k \cdot \nabla C_M^k$ is the diffusion flux vector of species k , and $\mathbf{D}^k$ is the diffusion coefficient tensor for species k . It should be pointed out that the drift term associated with the defect-sink interactions are implicitly incorporated into this CD framework by adopting the effective sink strength. This will be detailed in the next section. It is also worth noting that for the sake of having a similar reaction rate to the traditional rate theory, the above equations are written in the cluster density C_M , wherein each R_2^k is a 4×4 matrix, directly characterizing the interaction of the species k and all its counterparts. Assuming that the production and the reaction of these species are all homogeneous, the reaction and production matrices in Eq. (8) can be expressed as

$$R_2^v = - \begin{bmatrix} 0 & \beta^{v,i} & \beta^{v,2i} & \beta^{v,3i} \\ \beta^{v,i} & 0 & 0 & 0 \\ \beta^{v,2i} & 0 & 0 & 0 \\ \beta^{v,3i} & 0 & 0 & 0 \end{bmatrix} \quad R_2^i = - \begin{bmatrix} 0 & \beta^{v,i} & -\beta^{v,2i} & 0 \\ \beta^{v,i} & 2\beta^{i,i} & \beta^{i,2i} & 0 \\ -\beta^{v,2i} & \beta^{i,2i} & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$R_2^{2i} = - \begin{bmatrix} 0 & 0 & \beta^{v,2i} & -\beta^{v,3i} \\ 0 & -\beta^{i,i} & \beta^{i,2i} & 0 \\ \beta^{v,2i} & \beta^{i,2i} & 0 & 0 \\ -\beta^{v,3i} & 0 & 0 & 0 \end{bmatrix} \quad R_2^{3i} = - \begin{bmatrix} 0 & 0 & 0 & \beta^{v,3i} \\ 0 & 0 & -\beta^{i,2i} & 0 \\ 0 & -\beta^{i,2i} & 0 & 0 \\ \beta^{v,3i} & 0 & 0 & 0 \end{bmatrix} \quad (9)$$

$$R_1 = - \begin{bmatrix} (\langle k_{L,v}^2 \rangle + \langle k_{d,v}^2 \rangle) \overline{D}^v & 0 & 0 & 0 \\ 0 & (\langle k_{L,i}^2 \rangle + \langle k_{d,i}^2 \rangle) \overline{D}^i & -2\alpha^{i,2i} & -\alpha^{i,3i} \\ 0 & 0 & (\langle k_{L,2i}^2 \rangle + \langle k_{d,2i}^2 \rangle) \overline{D}^{2i} + \alpha^{i,2i} & -\alpha^{i,3i} \\ 0 & 0 & 0 & (\langle k_{L,3i}^2 \rangle + \langle k_{d,3i}^2 \rangle) \overline{D}^{3i} + \alpha^{i,3i} \end{bmatrix} \quad (10)$$

$$P_M = G \begin{bmatrix} 1 \\ 1 - 2\varepsilon_{2i} - 3\varepsilon_{3i} \\ \varepsilon_{2i} \\ \varepsilon_{3i} \end{bmatrix}; \quad \text{and} \quad C_M = \begin{bmatrix} C_M^v \\ C_M^i \\ C_M^{2i} \\ C_M^{3i} \end{bmatrix} \quad (11)$$

where $\beta^{m,n}$ in Eq. (9) represents the reaction rates between species m and n . The plus sign indicates the production while the negative sign means the consumption. It is thus noted that total conservation of atoms requires that $R_2^i + 2R_2^{2i} + 3R_2^{3i} - R_2^v = 0$. The sink strengths $\langle k_{L,m}^2 \rangle$ and $\langle k_{d,m}^2 \rangle$ solely appear in the diagonal components of R_1 , and $\alpha^{m,n}$ denotes the dissociation rate of n by emitting a species m . The $\langle k_{d,m}^2 \rangle$ is the sink strength of network dislocations for mobile species m , and has a similar expression as the dislocation loop strength $\langle k_{L,m}^2 \rangle$. In the present work we focus on studying irradiation growth of α -Zr, where the network dislocation density is low compared to the loop density, so from here on we neglect the term $\langle k_{d,m}^2 \rangle$. The loop number density was found to be in the order of 10^{22} m^{-3} for a temperature (573 K) and dpa level (0.06) comparable to the initial conditions of our study (Jostons et al., 1977). For the observed average size of $\approx 10 \text{ nm}$, this corresponds to more than two orders of magnitude higher line density compared to the typical network dislocation density in annealed α -Zr (10^{10} – 10^{12} m^{-2}) (Rogerson, 1988). Lastly, ε_{2i} and ε_{3i} are the fraction of di- and tri-interstitial clusters formed from the radiation cascade. We will later develop a weak form of Eq. (8) that can be solved for the mobile species concentrations, C_M , by the FE method in finite 3D geometry with appropriate boundary conditions. On the other hand, we distinguish large immobile clusters with their defect types, i.e., interstitial-type $\langle a_1 \rangle$, $\langle a_2 \rangle$, $\langle a_3 \rangle$ loops and vacancy-type $\langle c \rangle$ loops (or bi-pyramid). Accordingly, these large clusters are characterized by the number density, their radius, and their concentration, namely $N^L = [N_c^L, N_{a_1}^L, N_{a_2}^L, N_{a_3}^L]^T$, $r^L = [r_c^L, r_{a_1}^L, r_{a_2}^L, r_{a_3}^L]^T$ and $C^L = [C_c^L, C_{a_1}^L, C_{a_2}^L, C_{a_3}^L]^T$.

To formulate an equation for loop evolution, we now create a diagonal diffusion matrix, $\mathbf{D}_d$, for the diffusion coefficients, and sink strength matrix, $\mathbf{K}$, as

$$\mathbf{D}_d = \begin{bmatrix} \overline{D}^v & 0 & 0 & 0 \\ 0 & \overline{D}^i & 0 & 0 \\ 0 & 0 & \overline{D}^{2i} & 0 \\ 0 & 0 & 0 & \overline{D}^{3i} \end{bmatrix} \quad \mathbf{K} = \begin{bmatrix} \langle k_{c\mathcal{L},v}^2 \rangle & \langle k_{c\mathcal{L},i}^2 \rangle & \langle k_{c\mathcal{L},2i}^2 \rangle & \langle k_{c\mathcal{L},3i}^2 \rangle \\ \langle k_{a_1\mathcal{L},v}^2 \rangle & \langle k_{a_1\mathcal{L},i}^2 \rangle & \langle k_{a_1\mathcal{L},2i}^2 \rangle & \langle k_{a_1\mathcal{L},3i}^2 \rangle \\ \langle k_{a_2\mathcal{L},v}^2 \rangle & \langle k_{a_2\mathcal{L},i}^2 \rangle & \langle k_{a_2\mathcal{L},2i}^2 \rangle & \langle k_{a_2\mathcal{L},3i}^2 \rangle \\ \langle k_{a_3\mathcal{L},v}^2 \rangle & \langle k_{a_3\mathcal{L},i}^2 \rangle & \langle k_{a_3\mathcal{L},2i}^2 \rangle & \langle k_{a_3\mathcal{L},3i}^2 \rangle \end{bmatrix} \quad (12)$$

where $\overline{D}^m = \det(\mathbf{D}^m)$ is the average diffusion coefficient of the species m . Each $\langle k_{I\mathcal{L},m}^2 \rangle$ is the partial sink strength of $I\mathcal{L}$ -immobile clusters to the mobile defect m , and their sum over $I\mathcal{L}$ is the total sink strength of immobile clusters to the mobile species $\langle k_{\mathcal{L},m}^2 \rangle = \langle k_{c\mathcal{L},m}^2 \rangle + \langle k_{a_1\mathcal{L},m}^2 \rangle + \langle k_{a_2\mathcal{L},m}^2 \rangle + \langle k_{a_3\mathcal{L},m}^2 \rangle$.

Omitting the loop nucleation at the steady state, the immobile defect cluster evolution can then be characterized by the absorption of mobile defects, simply from Eq. (5) as,

$$\frac{dc_{\mathcal{L}}^c}{dt} = \mathbf{K} \cdot \mathbf{D}_d \cdot \mathbf{B}_M \mathbf{C}_M = \mathbf{K} \cdot \mathbf{D}_d \cdot \mathbf{c}_M \quad (13)$$

We emphasize again that the diffusion coefficient in the above equations is an average value accounting for anisotropic diffusion. We now turn attention to the expressions of the above reaction rates and sink strengths using the average diffusion coefficient. The accurate interaction rate associated with isotropic or anisotropic diffusing species has been thoroughly studied by Woo (1988). Here we simply approximate all these rates as

$$\beta^{m,n} = \frac{A^{m,n}}{\Omega} (\overline{D}^m + \overline{D}^n) (r^m + r^n) \quad (14)$$

where r^m is the radius of the species m depending on its geometry, $A^{m,n}$ is a constant characterizing the interaction efficiency between species m and n , and Ω is the atomic volume. It should be pointed out that this expression is similar to the isotropic diffusion cases, and the anisotropic properties are accounted for by adopting the average diffusion coefficient and a scaling parameter. The dissociation rate can then be written as

$$\alpha^{m,n} = \beta^{m,n} \exp\left(-\frac{E_{(n-m)}^b}{k_B T}\right) \quad (15)$$

where $E_{(n-m)}^b$ denotes the binding energy for an $(n-m)$ -size cluster, k_B is the Boltzmann constant, and T is the temperature.

In cluster dynamics modeling, the sink strength is a parameter characterizing the mobile species absorption efficiency of sinks, which depends on the sink type, its geometry, and the diffusion property of mobile defects. The effects of including defect-sink interactions can be also incorporated into this framework through a bias factor. The effective sink strengths of $\langle a \rangle$ -type prismatic loops, $\langle c \rangle$ -type basal loops and vacancy bi-pyramid are simply

$$\langle k_{a\mathcal{L},m}^2 \rangle = Z_{a\mathcal{L},tot}^m 2\pi r^{a\mathcal{L}} N^{a\mathcal{L}} \quad (16)$$

$$\langle k_{c\mathcal{L},m}^2 \rangle = Z_{c\mathcal{L},tot}^m 2\pi r^{c\mathcal{L}} N^{c\mathcal{L}} \quad (17)$$

$$\langle k_{bp,m}^2 \rangle = \alpha_{bp} s_{bp} \pi r_{bp} N_{bp} \quad (18)$$

where α_{bp} is a parameter scaling the bi-pyramid strength, and s_{bp} is a factor characterizing the shape of the cluster, e.g., $s_{bp} = 4$ for voids. $Z_{a\mathcal{L},tot}^m$ and $Z_{c\mathcal{L},tot}^m$ are total bias factors of $\langle a \rangle$ and $\langle c \rangle$ loops for the mobile defect m . The bias factors considering the effects of the anisotropic diffusion, as well as the external applied stress have been extensively studied in the past years (Bullough and Willis, 1975; Woo and Goesele, 1983; Skinner and Woo, 1984; Woo, 1988; Carpentier et al., 2017). According to models proposed by Woo (1988) and Borodin and Ryazanov (1994), the total bias factors can be written as

$$Z_{a\mathcal{L},tot}^m(\sigma) = Z_{a\mathcal{L},m}^{DAD,0} \left(1 + \frac{\Delta Z_{a\mathcal{L},m}^{SIPA}(\sigma)}{Z_{a\mathcal{L},m}^{DAD,0}} \right) \quad (19)$$

$$Z_{c\mathcal{L},tot}^m(\sigma) = Z_{c\mathcal{L},m}^{DAD,0} \left(1 + \frac{\Delta Z_{c\mathcal{L},m}^{SIPA}(\sigma)}{Z_{c\mathcal{L},m}^{DAD,0}} \right) \quad (20)$$

where $\Delta Z_{\mathcal{L},m}^{SIPA}(\sigma)$ accounts for the effect of stress on the absorption efficiency, depending on the orientation of the dislocation, known as SIPA. Details about the SIPA effect and the bias factors are discussed in Appendix A. Apart from the stress effect, the intrinsic diffusional anisotropy of defects in non-cubic crystals will also strongly influence the absorption efficiency. This effect is characterized by $Z_{\mathcal{L},m}^{DAD,0}$, which yields $Z_{a\mathcal{L},m}^{DAD,0} = Z_m^0 \left(\frac{p^m + (p^m)^{-2}}{2} \right)$, and $Z_{c\mathcal{L},m}^{DAD,0} = Z_m^0 p^m$, with $p^m = (D_c^m/D_a^m)^{1/6}$ being a parameter characterizing the anisotropy of defect diffusion on the basal plane (Woo and Goesele, 1983; Woo, 1988), and Z_m^0 the absorption efficiency of an isotropic defect without stress. It should be noted that rather than a presumed bias factor for DAD in traditional cluster dynamics model, the SR-CD model has the DAD effect naturally incorporated from the tensorial description of the flux of

mobile species. As the present work focuses on the irradiation growth of zirconium without the applied stress, the overall sink strength for dislocation loops can be simplified to

$$\langle k_{a\mathcal{L},m}^2 \rangle = Z_m^0 (p^m + (p^m)^{-2}) \pi r^{a\mathcal{L}} N^{a\mathcal{L}} \quad (21)$$

$$\langle k_{c\mathcal{L},m}^2 \rangle = Z_m^0 p^m 2\pi r^{c\mathcal{L}} N^{c\mathcal{L}} \quad (22)$$

It is obvious that for isotropic diffusing defects, i.e., single vacancies and single interstitials, $p = 1$, that the expression recovers the traditional sink strength of dislocations, while for di- and tri-interstitials with $p < 1$, there is preferential absorption of interstitial loops lying on the prismatic plane with $\langle k_{c\mathcal{L},m}^2 \rangle < \langle k_{a\mathcal{L},m}^2 \rangle$. It is notable that the sink strength of vacancy clusters is interpolated from $\langle k_{bp,m}^2 \rangle$ to $\langle k_{c\mathcal{L},m}^2 \rangle$.

2.3. Mechanics of inelastic deformation

Now consider the deformation of an infinitesimal *material* volume of a monoatomic crystal, that is an infinitesimal volume containing a fixed number of atoms (fixed mass). In the material reference frame we label a coordinate X , and in the spatial frame a coordinate $\mathbf{x} = \varphi(X, t)$. The deformation gradient tensor is then $F := \partial \mathbf{x} / \partial X$. This can be multiplicatively decomposed as $F = F^E F^I$, to include the elastic deformation F^E as well as the inelastic deformation F^I . The gradient of the material velocity $\mathbf{v} = \partial \varphi / \partial t$ is additively decomposed into elastic and inelastic components as $L = \partial \mathbf{v} / \partial \mathbf{x} = L^E + L^I$. Three distinct processes contribute to the inelastic velocity gradient (Po et al., 2022): L^P , the plastic deformation by dislocation glide and climb motion; L^D , the deformation due to mass transport to boundaries (i.e. the flux of mobile defect clusters) (Nabarro et al., 1948; Herring, 1950); and L^V , the volumetric deformation, associated with volume change when atoms are moved from lattice sites to reside in clusters, or form Frenkel Pairs. Hence we have

$$L^I = L^V + L^D + L^P \quad (23)$$

The objective is now to associate each of the deformation rates to its corresponding physical process. Consider the relative volume $J^I = dV^I / dV_0$, which is related to the defects concentrations as follows:

$$\begin{aligned} J^I &= \frac{dV^I}{dV_0} = \det(F^I) \\ &= \frac{\sum_1^{N^V} n \Omega_{nv} \# \text{cluster}_{nv} + \Omega_0 (\# \text{atoms} - \sum_1^{N^I} n \# \text{cluster}_{ni}) + \sum_1^{N^I} n \Delta \Omega_{ni} \# \text{cluster}_{ni} + \sum_{\mathcal{L}} g^{\mathcal{L}} \Omega^{\mathcal{L}} \# \text{loop}}{\Omega_0 \# \text{atoms}} \\ &= 1 + \sum_{nv} \left(1 - \frac{\Delta \Omega_0^{nv}}{\Omega_0} \right) n C_M^{nv} - \sum_{ni} \left(1 - \frac{\Delta \Omega_0^{ni}}{\Omega_0} \right) n C_M^{ni} + \sum_{\mathcal{L}} g^{\mathcal{L}} (b^{\mathcal{L}} \cdot A^{\mathcal{L}}) N^{\mathcal{L}} \\ &= 1 - \sum_n \left(1 - \frac{\Delta \Omega_0^n}{\Omega_0} \right) c_M^n + \sum_{\mathcal{L}} g^{\mathcal{L}} (b^{\mathcal{L}} \cdot A^{\mathcal{L}}) N^{\mathcal{L}} \\ &= 1 - \sum_n \left(1 - \frac{\Delta \Omega_0^n}{\Omega_0} \right) c_M^n + \sum_{\mathcal{L}} g^{\mathcal{L}} c_I^{\mathcal{L}} \end{aligned} \quad (24)$$

where $c_M^n := n \# \text{defect} / \# \text{atoms}$ is the number density for the size- n mobile cluster defined in Eq. (2) and $g^{\mathcal{L}}$ is the relative relaxation volume associated with a dislocation loop.

From Eq. (24), the rate of J^I is

$$\begin{aligned} \dot{J}^I &= J^I \text{tr}(\dot{L}^I) = J^I \text{tr}(L^I) = J^I [\text{tr}(L^P) + \text{tr}(L^D) + \text{tr}(L^V)] \\ &= - \sum_n \left(1 - \frac{\Delta \Omega_0^n}{\Omega_0} \right) \dot{c}_M^n + \sum_{\mathcal{L}} g^{\mathcal{L}} \dot{c}_I^{\mathcal{L}} \end{aligned} \quad (25)$$

Now, under steady-state conditions with no cascade production, plugging the defect concentration balance laws into Eq. (25) and applying the conservation of atoms yields an expression relating the deformation rates to their physical processes. Since the atomic volume in the stressed crystal is $\Omega = J^E \Omega_0$, then

$$J^I [\text{tr}(L^P) + \text{tr}(L^D) + \text{tr}(L^V)] = \sum_n \nabla \cdot \mathbf{J}_M^n + \sum_n \frac{\Delta \Omega_0^n}{\Omega_0} \dot{c}_M^n + \sum_{\mathcal{L}} (1 + g^{\mathcal{L}}) \dot{c}_I^{\mathcal{L}} \quad (26)$$

Eq. (26) is the fundamental kinematic relation that allows to associate the rates of deformation tensors L^P , L^V , and L^D to the corresponding physical processes. Since the defect loops are sources of the dislocation climb process, then

$$J^I \text{tr}(L^P) = \sum_{\mathcal{L}} \dot{c}_I^{\mathcal{L}} = \sum_{\mathcal{L}} \frac{d}{dt} ((b^{\mathcal{L}} \cdot A^{\mathcal{L}}) N^{\mathcal{L}}) \quad \text{or} \quad L^P = \sum_{\mathcal{L}} \frac{d}{dt} ((b^{\mathcal{L}} \otimes A^{\mathcal{L}}) N_I^{\mathcal{L}}) \quad (27)$$

Note that Eq. (27) neglects the contribution of network dislocations, which is negligible under growth conditions, that is in the absence of an applied stress. This is because the dislocation network density is negligible compared to the loop density, as already

discussed in relation to Eq. (10). Next, the fact that F^V is the volumetric deformation induced by the defect relaxation volume suggests that

$$J^I \text{tr}(\mathbf{L}^V) = \sum_n \frac{\Delta\Omega_0^n}{\Omega_0} \dot{c}_M^n + \sum_{\mathcal{L}} f^{\mathcal{L}} \dot{c}_I^{\mathcal{L}} \quad \text{or} \quad \mathbf{L}^V = \sum_n \frac{\Delta\Omega_0^n}{3\Omega_0} \dot{c}_M^n \mathbf{I} + \sum_{\mathcal{L}} f^{\mathcal{L}} \frac{d}{dt} ((b^{\mathcal{L}} \otimes \mathbf{A}^{\mathcal{L}}) N_I^{\mathcal{L}}) \quad (28)$$

Finally, we are left with the condition $\text{tr}(\mathbf{L}^D) = (\sum_n \nabla \cdot \mathbf{B}_M^n \mathbf{J}_M^n)$, which is satisfied by the following definition of the vacancy diffusion velocity “gradient”:

$$J^I \text{tr}(\mathbf{L}^D) = \sum_n \nabla \cdot \mathbf{J}_M^n \quad \text{or} \quad \mathbf{L}^D = \sum_n \nabla \mathbf{J}_M^n \quad (29)$$

$\mathbf{L}^D$ can also describe the bulk diffusion of species to the boundaries, and this type of creep can be dominant at low stresses and high temperatures. Then, combining Eqs. (27) and (13) the strain rate tensor is simply computed as

$$\begin{aligned} \mathbf{L}^P = & \left\langle k_{a_1 \mathcal{L}, m}^2 \right\rangle \overline{\mathbf{D}}^m c_M^m (\hat{\mathbf{a}}_1 \otimes \hat{\mathbf{a}}_1) + \left\langle k_{a_2 \mathcal{L}, m}^2 \right\rangle \overline{\mathbf{D}}^m c_M^m (\hat{\mathbf{a}}_2 \otimes \hat{\mathbf{a}}_2) \\ & + \left\langle k_{a_3 \mathcal{L}, m}^2 \right\rangle \overline{\mathbf{D}}^m c_M^m (\hat{\mathbf{a}}_3 \otimes \hat{\mathbf{a}}_3) + \left\langle k_{c \mathcal{L}, m}^2 \right\rangle \overline{\mathbf{D}}^m c_M^m (\hat{\mathbf{c}} \otimes \hat{\mathbf{c}}) \end{aligned} \quad (30)$$

where $\hat{\mathbf{a}}_1, \hat{\mathbf{a}}_2, \hat{\mathbf{a}}_3$ and $\hat{\mathbf{c}}$ are unit lattice vectors in the global Cartesian system. Swelling is an isotropic material volume expansion attributed to volumetric part of the total inelastic deformation tensor $\mathbf{L}^I$. Under loading, the deviatoric part of $\mathbf{L}^I$ is the creep rate tensor and the volumetric part is the swelling tensor. However, with no loading, the deviatoric part of $\mathbf{L}^I$ is the growth rate tensor. The swelling and growth rate tensors are expressed as

$$\dot{\epsilon}_{SW} = \frac{1}{3} \text{tr}(\mathbf{L}^I) \cdot \mathbf{I} \quad \dot{\epsilon}_{GR} = \mathbf{L}^I - \frac{1}{3} \text{tr}(\mathbf{L}^I) \cdot \mathbf{I} \quad (31)$$

where $\mathbf{I}$ is the identity matrix. Eqs. (28), (29), and (30) are kinematic relations needed to develop the irreversible thermodynamics of the defect microstructure evolution process.

3. Finite element solution method

3.1. Weak form of the SR-CD equations & boundary conditions

Each density is subject to Dirichlet boundary conditions, where the prescribed value of the concentration depends on the point-wise mechanical traction on the boundary. For example, the boundary value (BND) of the vacancy concentration is (Po et al., 2022)

$$C^v = C_{\text{BND}}^v := C_0^v \exp\left(\frac{g_v + t_n \Omega_0}{k_B T}\right) \quad \text{on } \partial\mathcal{V} \quad (32)$$

where g is the formation energy per vacancy on the boundary (typically zero for external boundaries), t_n the normal traction to the boundary, and Ω_0 the atomic volume. Moreover, $C_0^v = \exp\left(-\frac{E_{vf} + \frac{\Delta\Omega_{0v}}{3} \text{tr}(\sigma)}{k_B T}\right)$ is the equilibrium vacancy concentration at the local temperature and stress, while E_{vf} is the vacancy formation energy, and $\Delta\Omega_{0v}$ is the vacancy relaxation volume. Similar boundary conditions apply to the other defect densities, with parameters specific to each. This non-linear problem is solved using a full-Jacobian Newton–Raphson scheme. The initial solution of the Newton–Raphson iteration is obtained by solving the boundary value problem without the reaction terms. Therefore, the initial solution $\tilde{C}_0$ satisfies the weak BVP:

$$\int_V \hat{C}^\alpha (\nabla \cdot \tilde{\mathbf{J}}_0^\alpha + P^\alpha) dV = 0 \quad \forall \hat{C}^\alpha. \quad (33)$$

Which can be written in shorthand notation, using the L^2 inner product as

$$\left\langle \hat{C}^\alpha, -\nabla \cdot \tilde{\mathbf{J}}_0^\alpha + P^\alpha \right\rangle_{L^2} = 0 \quad \forall \hat{C}^\alpha. \quad (34)$$

Applying the divergence theorem, this becomes

$$-\left\langle \nabla \hat{C}^\alpha, \tilde{\mathbf{J}}_0^\alpha \right\rangle = \left\langle \hat{C}^\alpha, P^\alpha \right\rangle \quad \forall \hat{C}^\alpha. \quad (35)$$

Once $\tilde{C}_0$ is obtained solving (35), we apply the Newtown–Raphson scheme to obtain the solution with non-linear reactions. To this end, we use an iterative solution scheme. We update the concentrations in iteration $k+1$ from those in k as $\tilde{C}_{k+1}^\alpha = \tilde{C}_k^\alpha + d\tilde{C}^\alpha$. We solve for $d\tilde{C}_k^\alpha$ at each iteration with *homogeneous* BCs over the whole domain. This requires the solution of the weak BVP

$$\left\langle \hat{C}^\alpha, -\nabla \cdot (\tilde{\mathbf{J}}_k^\alpha + d\tilde{\mathbf{J}}^\alpha) + \frac{1}{2} R_2^{\alpha\beta\gamma} \tilde{C}_k^\beta \tilde{C}_k^\gamma + R_2^{\alpha\beta\gamma} \tilde{C}_k^\gamma d\tilde{C}^\beta + R_1^{\alpha\beta} \tilde{C}_k^\beta + R_1^{\alpha\beta} d\tilde{C}^\beta + P_k^\alpha \right\rangle = 0 \quad \forall \hat{C}^\alpha \quad (36)$$

Applying the divergence theorem, this yields

$$-\left[\left\langle \nabla \hat{C}^\alpha, d\tilde{\mathbf{J}}^\alpha \right\rangle + \left\langle \hat{C}^\alpha, R_2^{\alpha\beta\gamma} \tilde{C}_k^\gamma d\tilde{C}^\beta + R_1^{\alpha\beta} d\tilde{C}^\beta \right\rangle \right] = \left\langle \nabla \hat{C}^\alpha, \tilde{\mathbf{J}}_k^\alpha \right\rangle + \left\langle \hat{C}^\alpha, P_k^\alpha \right\rangle$$

$$+ \left\langle \hat{C}^\alpha, \frac{1}{2} R_2^{\alpha\beta\gamma} \tilde{C}_k^\beta \tilde{C}_k^\gamma + R_1^{\alpha\beta} \tilde{C}_k^\beta \right\rangle \quad \forall \hat{C}^\alpha$$

A flowchart illustrating the CD-mechanics coupled scheme is presented in Fig. 2, which is divided into three sections by color: inputs (beige), solution and updating steps (blue), and post processing (pink). The mechanics boundary value problem (BVP) is initially solved to determine the displacement and stress state based on the specified boundary conditions. Subsequently, the stress state, initial densities of immobile species, and concentrations of immobile species are used to compute their sink strengths. With these values, R_1 and R_2 can be constructed, and together with the boundary conditions from the mechanics BVP, the BVP for the concentrations of mobile species can be solved. During this step, the Newton Raphson scheme is employed until the solution to the boundary valued problem has converged. Upon obtaining the concentrations of mobile species, the growth rate of immobile species can be computed, and their concentrations are accordingly updated. It is assumed that there is no nucleation of immobile defects. Simultaneously, the diffusive flux $\mathbf{J}_M$ can be computed from the concentrations of mobile species, and the diffusive displacement is then updated after solving the BVP for dC_k . Additionally, the plastic distortion and volumetric distortion are updated within the mechanics BVP, as they influence the stress state, which in turn affects the concentrations of mobile species. The volumetric deformation from mobile species is typically considered negligible compared to the strain from dislocation loops.

3.2. SR-CD model parameters

The computational model is based on a number of parameters, and the accuracy of the results will naturally depend on the precision of these parameters. A full list of model parameters for the baseline simulation fit to the experimental data for single crystals from Carpenter et al. (1988), is shown in Table 1. The numerical implementation is in a Cartesian frame, and lattice vectors can be easily converted to the Miller–Bravais representation as: $\mathbf{a}_1 = ae_x$, $\mathbf{a}_2 = -\frac{1}{2}ae_x + \frac{\sqrt{3}}{2}ae_y$, $\mathbf{a}_3 = -\frac{1}{2}ae_x - \frac{\sqrt{3}}{2}ae_y$, $\mathbf{c} = ce_z$. A hexagonal cell with $c/a = \sqrt{8/3}$ is adopted. The magnitude of burgers vector for $\langle c \rangle$ loops is taken as $|\mathbf{b}^c| = \frac{c}{a} \times 0.3233 \times 10^{-8}$ cm, and for $\langle a \rangle$ loops is taken as $|\mathbf{b}^a| = 0.3233 \times 10^{-8}$ cm (Griffiths et al., 1983; Li et al., 2020). Following typical operating conditions for LWRs, the simulation temperature and irradiation dose were 553 K and 1×10^{-7} dpa/s, respectively (Carpenter et al., 1988; Adamson et al., 2019).

Defect energies are sourced from experiments or empirical potentials for hcp zirconium. Notably, and the Mendelev and Ackland #3 potential is the most commonly used interatomic potential for Zr (Mendelev and Ackland, 2007). Experimental evidence suggests that the formation energy of vacancies is approximately 1.5 eV, although empirical potentials provide a range of 1.7–2.3 eV (Ouyang et al., 2018; Mikhin et al., 1994; Hu et al., 2001; Dai et al., 2009). Vacancy migration exhibits slight anisotropy, with higher energies in the basal plane (0.67–1.14 eV) compared to out-of-plane migration (0.72–0.89 eV) according to empirical potentials (Ouyang et al., 2018; Mikhin et al., 1994; Kim et al., 2006; Hu et al., 2001; Dai et al., 2009). For experimental data, the migration energy in the basal plane ranges from 0.65–0.98 eV, and out of the basal plane was found in one case to be 1.02 eV (Hood, 1986; Buckley et al., 1980; Horvath et al., 1984). Interpolation between experimental and empirical data establishes a migration energy of 0.9 eV for vacancies in all directions. Meanwhile, the diffusion coefficient for vacancies is derived from prior research (Li et al., 2020).

Interstitial defects exhibit formation energies of 3.0 eV for the most stable types, and a migration energy of 0.09 eV is taken in all directions (Mikhin et al., 1994; Ouyang et al., 2018). The most stable site for interstitials, identified through empirical potentials, is the octahedral or basal-octahedral site (Peng et al., 2012; Ouyang et al., 2018). A relative relaxation of 20% is assumed for both interstitials and larger mobile defects (di and tri). Large cluster properties demonstrate an inverse relationship between formation and binding energies with increasing size (De Diego et al., 2002). Di-interstitials possess a formation energy of 6 eV and a binding energy of 1.42 eV (Christien and Barbu, 2005), while tri-interstitials exhibit a formation energy of 8 eV and a binding energy of 4 eV (De Diego et al., 2002). Both di- and tri-interstitials display similar but faster mobility to single interstitials with comparable basal-plane migration energies for various c/a ratios (Mikhin et al., 1994). Out of the basal plane, the perfect c/a (1.6) was found to behave isotropically, while the experimental c/a (1.58) behaved anisotropically (Mikhin et al., 1994). Migration energies in the basal plane for these defects are set at 0.01 eV, with out-of-plane migration energies at 0.045 eV. The diffusion coefficient for all interstitial species was also taken following our previous work (Li et al., 2020).

For immobile clusters, the relative relaxation volumes tend to approach $\lim_{N \rightarrow \infty} \frac{\Omega_{rel}}{\Omega_0} = \pm N$, where N is the number of defects in the dislocation loop (Mason et al., 2019), and where a positive or negative indicates interstitial or vacancy loops, respectively. This limit is almost reached at a size of 1000 vacancies or interstitials, where the vacancy clusters are still transitioning from bi-pyramids into full dislocation loops. The relaxation volumes of vacancy loops and interstitial loops are taken as 10% and 1%, respectively. Atomic simulations confirm the stability of vacancy bi-pyramids (Voskoboinikov et al., 2006), with a stability transition occurring between bi-pyramids and perfect $\langle c \rangle$ dislocation loops at a critical cluster size (Christiaen et al., 2019). Here, this transition is observed at approximately 400 vacancies, progressing to prismatic $\langle c \rangle$ dislocation loops at 1500 vacancies. The change in material volume due to the eigenstrain of vacancy clusters is determined from atomic simulations, slightly adjusted to 5×10^{-26} m³ per cluster (Christiaen et al., 2019). There is no bias associated with immobile vacancy clusters according to Eq. (18), while the interaction bias of all mobile species with dislocations is assumed to be 1.0. The sink strength coefficient for bi-pyramid clusters, relative to the sink strength of a void, is set at 40% to match the initial experimental growth strain.

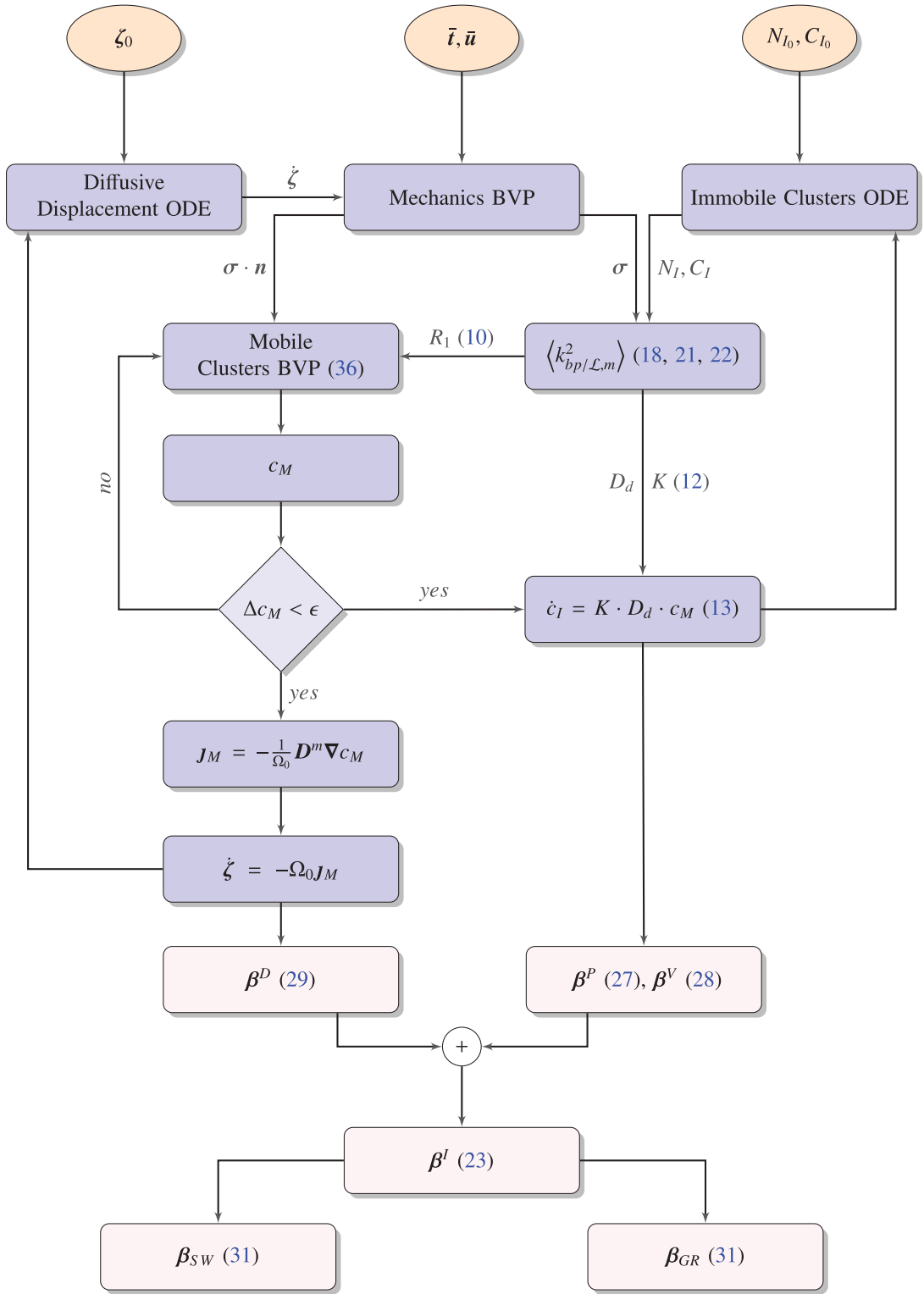


Fig. 2. Flow chart of the simulation framework. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1

Model parameters used in reduced-CD simulations for the irradiation damage production, and mobile and immobile species properties.

Definition	Symbol	Value	Reference/Comment
Damage production			
Temperature	T	553 K	Carpenter et al. (1981, 1988)
Dose rate	G_0	10^{-7} dpa/s	Carpenter et al. (1981, 1988)
Fraction of recombining point defects	f_r	0.99	Zhou et al. (2018) and Kim et al. (2018)
Fraction of total interstitials produced as single interstitials	ε_i	0.6	Zhou et al. (2018)
Fraction of total interstitials produced as di-interstitial clusters	ε_{2i}	0.16	Bacon et al. (2003)
Fraction of total interstitials produced as tri-interstitial clusters	ε_{3i}	0.026	Bacon et al. (2003)
Mobile species			
Relative relaxation volume of vacancy	$\Delta\Omega_v$	0.405	Varvenne et al. (2014)
Relative relaxation volume of single, di-, and tri-interstitial	$\Delta\Omega_{i,2i,3i}$	1.2	Li et al. (2020)
Formation energy of vacancy	E_v^f	1.5 eV	Ouyang et al. (2018) and Hood (1986)
Formation energy of interstitial	E_i^f	3.0 eV	Ouyang et al. (2018) and Peng et al. (2012)
Formation energy of di-interstitial	E_{2i}^f	6.0 eV	De Diego et al. (2002)
Formation energy of tri-interstitial	E_{3i}^f	8.0 eV	De Diego et al. (2002)
Binding energy of di-interstitial	E_{2i}^b	1.42 eV	Christien and Barbu (2009)
Binding energy of tri-interstitial	E_{3i}^b	4.0 eV	De Diego et al. (2002)
Migration energies of vacancy	$E_v^{(11,22,33)}$	0.9 eV	Ouyang et al. (2018)
Migration energies of interstitial	$E_i^{(11,22,33)}$	0.09 eV	Mikhin et al. (1994)
Migration energies of di- and tri-interstitial	$E_{2i,3i}^{(11,22)}$	0.01 eV	Mikhin et al. (1994)
Migration energy of di- and tri-interstitial	$E_{2i,3i}^{(33)}$	0.045 eV	Mikhin et al. (1994)
Diffusion coefficient of vacancy	D_v	4.9639×10^{-10} m ² /s	Li et al. (2020)
Diffusion coefficient of single, di-, and tri-interstitials	$D_{i,2i,3i}$	8.1882×10^{-10} m ² /s	Li et al. (2020)
Interaction bias factor of vacancies	Z_v^0	1.0	Woo (1988) and Wolfer (2007)
Interaction bias factor of single, di- and tri-interstitials	$Z_{i,2i,3i}^0$	1.0	Woo (1988) and Wolfer (2007)
Immobile species			
Relative relaxation volume of vacancy loop	$g^{(c)}$	0.1	Mason et al. (2019)
Relative relaxation volume of interstitial loop	$g^{(a_1,a_2,a_3)}$	0.01	Mason et al. (2019)
Initial $\langle c \rangle$ number of loop defects per lattice cite	$C_l^{(c)}$	2×10^{-5}	Li et al. (2020)
Initial $\langle a_1, a_2, a_3 \rangle$ number of loop defects per lattice cite	$C_l^{(a_1,a_2,a_3)}$	12×10^{-5}	Li et al. (2020)
Initial $\langle c \rangle$ vacancy loop number per simulation Volume	$N_l^{(c)}$	4×10^{21} L/m ³	Li et al. (2020)
Initial $\langle a_1, a_2, a_3 \rangle$ interstitial loop number per simulation Volume	$N_l^{(a_1,a_2,a_3)}$	4×10^{21} L/m ³	Li et al. (2020)
Critical number of vacancies below which cluster is a bi-pyramid	n_{min}	400	Christiaen et al. (2019)
Critical number of vacancies above which cluster is a loop	n_{max}	1500	Christiaen et al. (2019)
Change in volume of a bi-pyramid defect for each cluster	$\Delta\Omega_{bp}$	5.0×10^{-26} m ³	Christiaen et al. (2019)
Bi-pyramid sink strength coefficient	α_{bp}	0.40	Adjusted

4. Results

4.1. Irradiation growth in Zr single crystals

The experimental data of Carpenter et al. (1981), for both iodide and zone refined Zr are plotted in Fig. 3(a), alongside the model results. A single crystal simulation of a 1- μ m size is compared to the experimental growth data. A study on mesh size convergence has been included in Appendix B. The current simulation results show distinct stages II and stage III, where growth behavior changes from steady-state to breakaway. Fig. 3(b) shows the corresponding vacancy bi-pyramid to vacancy loop transition function, where between 400 and 1500 vacancies, clusters interpolated bi-pyramid/loop defects. The reader is reminded of the sigmoidal interpolation function used to model properties of vacancy clusters in Section 2.2. To visualize the resulting microstructure, including the bi-pyramid-to-loop transition, the FE fields are plotted as discrete loops and bi-pyramids, shown in Fig. 4, by selecting random locations in the material volume. The growth strains along both the $\langle a \rangle$ axis and $\langle c \rangle$ axis are in excellent agreement with the experimental data of single crystal Zr (Carpenter et al., 1981).

Between 1–3 dpa the vacancy bi-pyramids act to reduce the growth strains along the $\langle a \rangle$ axis of the crystal as their isotropic eigenstrains compensate the $\langle a \rangle$ axis expansion. This causes the steady-state growth experienced by the crystal during stage II. As neutron damage increases, the overall contribution to the $\langle a \rangle$ axis strain from the bi-pyramids approaches zero as they collapse into loops, shown by the blue dash-dot curve in Fig. 3(a). The $\langle a \rangle$ axis begins to display slight breakaway behavior at 3 dpa. Referring to Fig. 4, $\langle c \rangle$ loops begin to form in the simulation volume at 3 dpa. Experiments by Holt and Gilbert (1986), postulated that $\langle c \rangle$ dislocation loop formation coincided with the breakaway growth at 3 dpa experienced by Carpenter et al. (1981). More recent TEM experiments have also shown that not until 3 dpa, can $\langle c \rangle$ dislocation loops be found (Harte et al., 2017; Onimus et al., 2020b).

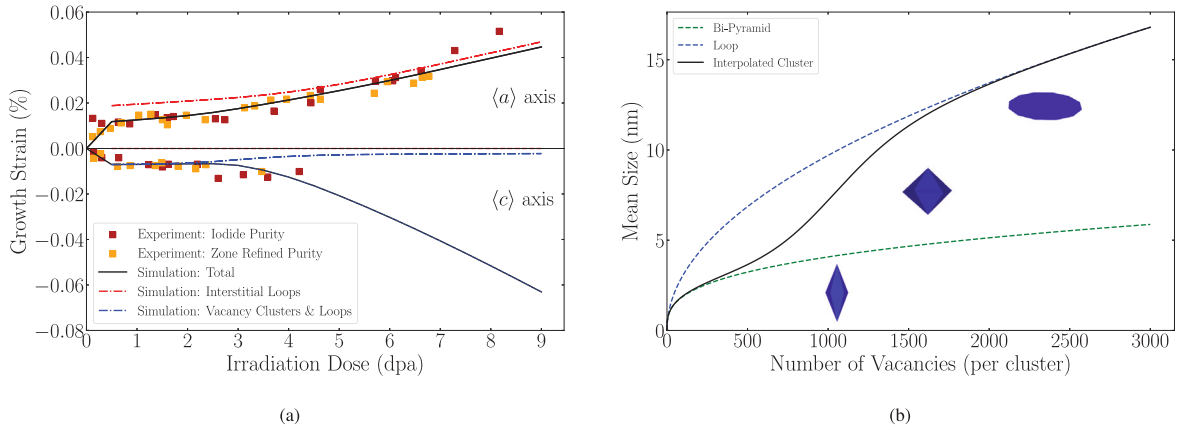


Fig. 3. (a) Simulation results compared to Carpenter and Rogerson experimental growth strain data (Carpenter et al., 1981). Dashed-dot curves show contributions from interstitial clusters (red) and vacancy clusters (blue) to the overall $\langle a \rangle$ axis growth strain. Dashed curves similarly show their contributions to the overall $\langle c \rangle$ axis growth strain. (b) Vacancy bi-pyramid to vacancy loop transition function for the radius of a cluster. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

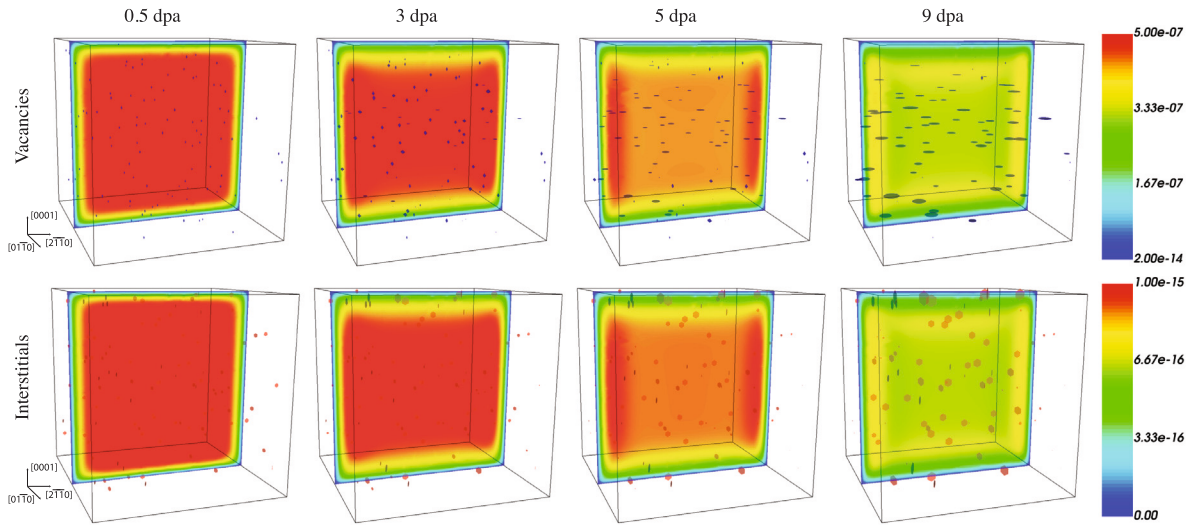


Fig. 4. Field plots are shown for mobile defect concentrations: vacancy and di-interstitials. Also displayed are the spatially resolved distributions of immobile species: basal $\langle c \rangle$ bi-pyramids and vacancy loops (blue), and prismatic $\langle a \rangle$ interstitial loops (red) for visualization only (randomly selected locations, discretized, and scaled $\times 2$). Irradiation doses shown are 0.5, 3, 5, and 9 dpa, highlighting the correlation between $\langle c \rangle$ loop formation and corresponding breakaway growth shown in Fig. 3. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Since interstitial loops are planar defects on the prismatic planes, they have no contribution to the total $\langle c \rangle$ axis strain, as shown by the red dashed curve. Therefore, the only contribution to the growth strain along the $\langle c \rangle$ axis is from vacancy bi-pyramids and vacancy loops, and is the same as the total $\langle c \rangle$ axis strain. After bi-pyramids collapse, the $\langle c \rangle$ dislocation loops have anisotropic eigenstrains, and begin to grow rapidly as their sink strength has increased out of the basal plane, resulting in the breakaway along both crystallographic axes. There is always a high interstitial defect flux, and interstitial loops continuously grow, but slowly, thus the breakaway growth during stage III is directly correlated with the formation of $\langle c \rangle$ dislocation loops. The vacancy cluster to vacancy loop transition was, again, postulated by Holt and Gilbert (1986), as a potential nucleation mechanism of $\langle c \rangle$ vacancy loops, and a potential cause of breakaway growth. In the time since Holt's hypothesis, the stability of the bi-pyramid like vacancy clusters have been demonstrated in atomic simulations (Bacon et al., 2003; Voskoboinikov et al., 2006), and the switch of the most stable defect to a $\langle c \rangle$ dislocation loop has also been shown (Christiaen et al., 2019). Furthermore, recent experiments have found similar triangular volumetric defects that transition into dislocation loops, after an incubation period (Liu et al., 2020). With SR-CD we have shown that the eigenstrain transformation of isotropic $\langle c \rangle$ vacancy clusters into anisotropic $\langle c \rangle$ vacancy loops is the phenomenon responsible for both breakaway growth and steady-state growth, and have agreement with experimental observations for irradiation induced growth strains, and for the dose where breakaway occurs.

Table 2Mean size and density for $\langle a \rangle$ and $\langle c \rangle$ dislocation loops in Zr and Zr-alloys irradiated at different temperatures^a and particle fluence/dpa.

Alloy	Irradiation particle	Equivalent dpa	Temperature	Loop density	Mean size	Reference
$\langle a \rangle$ loops						
Zr-2	Neutron	0.52 dpa	523 K	$3.4 \times 10^{22} \text{ } \mathcal{L}/\text{m}^3$	8.5 nm	Northwood et al. (1979)
Zr	Neutron	0.5, 10, 20 dpa	553 K	$1.2 \times 10^{22} \text{ } \mathcal{L}/\text{m}^3$	10.9, 11.3, 15.1 nm	Model results
Zr-2	Neutron	15, 25 dpa	573 K	$1, 1.5 \times 10^{22} \text{ } \mathcal{L}/\text{m}^3$	5 nm	Harte et al. (2017)
Zr	Neutron	<0.1 dpa ^b	573 K	$1.6 \times 10^{21} \text{ } \mathcal{L}/\text{m}^3$	9 nm	Jostsons et al. (1977)
Zr	Neutron	0.8 dpa	573 K	$5.0 \times 10^{21} \text{ } \mathcal{L}/\text{m}^3$	21 nm	Gilbert et al. (1979)
Zr-2	Neutron	4.8, 12, 23 dpa	600 K	$4.3, 4.9, 5.1 \times 10^{22} \text{ } \mathcal{L}/\text{m}^3$	10, 10.5, 10.7 nm	Kobylansky et al. (2008)
NSF	Neutron	23 dpa	600 K	$8.8 \times 10^{22} \text{ } \mathcal{L}/\text{m}^3$	6.6 nm	Kobylansky et al. (2008)
E635	Neutron	23 dpa	600 K	$7.4 \times 10^{22} \text{ } \mathcal{L}/\text{m}^3$	7.3 nm	Kobylansky et al. (2008)
Zr-2	Neutron	1.67 dpa	615 K	$1.9 \times 10^{22} \text{ } \mathcal{L}/\text{m}^3$	8.7 nm	Northwood et al. (1979)
Zr	Neutron	<0.1 dpa ^b	623 K	$8.0 \times 10^{20} \text{ } \mathcal{L}/\text{m}^3$	<10 nm	Jostsons et al. (1977)
$\langle c \rangle$ loops						
Zr	Neutron	0.5, 10, 20 dpa	553 K	$4 \times 10^{21} \text{ } \mathcal{L}/\text{m}^3$	0, 22, 35 nm	Model results
Zr-2	Neutron	15, 25 dpa	573 K	$4, 8 \times 10^{20} \text{ } \mathcal{L}/\text{m}^3$	100 nm	Harte et al. (2017)
Zr-4	Zr Ion	4.1, 5.5, 7 dpa	573 K	$2.9, 6.7, 9.1 \times 10^{20} \text{ } \mathcal{L}/\text{m}^3$	24, 27, 36 nm	Tournadre et al. (2012)
Zr-4	Neutron	14, 35 dpa	583 K	$1.2, 1.17 \times 10^{21} \text{ } \mathcal{L}/\text{m}^3$	>150 nm	Tournadre et al. (2012)

^a The size and density of dislocation loops during irradiation is highly dependent on temperature (Adamson et al., 2019), and experiments have shown loop density decreases, but mean size increases, with increasing irradiation temperature (Hellio et al., 1988; Gaumé et al., 2017; Gilbert et al., 1979).

^b Conversion to dpa: 1 dpa = $0.6 \times 10^{25} \text{ n/m}^2$ (Carpenter et al., 1981; Adamson et al., 2019).

Upon inspecting the spatially-resolved results, it becomes apparent that the two immobile defect species grow independently in different spatial regions, as illustrated by the microstructure at 9 dpa in Fig. 4. The preferential growth of $\langle a \rangle$ loops near boundaries parallel to the [0001] basal pole direction, and $\langle c \rangle$ loops perpendicular to this direction, is attributed to the disparity in boundary layer size along the $\langle a \rangle$ and $\langle c \rangle$ axes, caused by the diffusion anisotropic difference (DAD). Along the basal pole, di- and tri-interstitials diffuse more slowly, resulting in a larger vacancy boundary layer compared to the interstitial layer, thereby promoting $\langle a \rangle$ loop growth. Conversely, boundaries parallel to the basal pole exhibit a larger relative interstitial boundary layer, favoring the preferential growth of $\langle c \rangle$ loops. These spatially-dependent growth patterns of basal $\langle c \rangle$ dislocation loops and prismatic $\langle a \rangle$ dislocation loops elucidated with SR-CD could potentially explain certain TEM findings. For instance, in some TEM experiments, locations where $\langle a \rangle$ loops were found $\langle c \rangle$ loops were not, and vice versa (Holt and Gilbert, 1986). Experiments by Harte on proton and neutron irradiated Zr alloys also provide direct evidence of “non-correlated” growth positions of the $\langle a \rangle$ and $\langle c \rangle$ loops (Harte et al., 2017). Other experiments conducted by Gilbert et al. (1979) have shown denuded zones near grain boundaries, and our findings suggest that interstitial loops grow more rapidly near specific boundaries depending on crystal orientation. The interactions between grain boundaries and dislocation loops are not addressed here and remain a subject for future investigation, however, such interactions could potentially lead to the absorption of large dislocation loops, leaving behind the denuded zones. Notably, a Kr ion-irradiated nanocrystalline Ni experiment observed the climb and absorption of individual dislocation loops into a grain boundary (Sun et al., 2013). Furthermore, dislocation channeling observed along different directions in polycrystalline Zirconium alloys, as well as strain localization, could be born from the independent spatial growth phenomenon between dislocation loop species as observed here (Onimus et al., 2020a; McMurtrey et al., 2014). A strong dependence of channeling and irradiation induced hardening on grain boundaries, grain orientation, and irradiation induced defect densities has also been observed (Erinosho and Dunne, 2015; Wang et al., 2020; Liu et al., 2022).

It is noteworthy that high concentrations of $\langle c \rangle$ vacancy loops only emerge once breakaway growth begins at doses exceeding 3 dpa or $3 - 4 \times 10^{25} \text{ n/m}^2$, as evidenced in irradiated Zr experiments (Kobylansky et al., 2008; Onimus et al., 2020b; Harte et al., 2017; Holt and Gilbert, 1983). Irradiation doses below this threshold have consistently produced only $\langle a \rangle$ component dislocation loops (Northwood et al., 1979). Breakaway growth, associated with the formation of $\langle c \rangle$ loops, implies an incubation period linked to the growth of vacancy clusters. Jostsons’ experiments (Jostsons et al., 1977) on zone-refined Zr irradiated to $6.4 \times 10^{23} \text{ n/m}^2$, less than 1 dpa, revealed the presence of $\langle a \rangle$ dislocation loops in certain micrographs, whereas $\langle c \rangle$ dislocation loops were absent in the same region. Holt’s experiments at higher irradiation doses showed the coexistence of $\langle c \rangle$ and $\langle a \rangle$ dislocation loops, however, sometimes a high density of $\langle c \rangle$ dislocation loops was found, sometimes a low density, and sometimes a non uniformly distributed density, particularly around precipitates, was observed (Holt and Gilbert, 1986). Similarly, other studies have documented preferential nucleation of $\langle c \rangle$ loops around precipitates (Onimus et al., 2020b; Kobylansky et al., 2008, 2011).¹ At low dose, $\langle c \rangle$ component loops are not found, and at high irradiation doses, matrix chemistry and precipitation can cause a spatial variation of $\langle c \rangle$ dislocation loops. Our spatially-resolved cluster dynamics (SR-CD) findings reveal an incubation period for vacancy cluster growth and a spatial variation in the growth of $\langle a \rangle$ and $\langle c \rangle$ dislocation loops, due to the boundary sinks within the irradiated single crystal volume.

¹ Alloying effects significantly influence irradiation damage evolution; for instance, niobium addition has experimentally demonstrated a reduction in growth strains and delayed breakaway growth (Kobylansky et al., 2008; Adamson et al., 2019), along with induced precipitation during irradiation (Onimus et al., 2020a,b). Additionally, experiments have shown that iron reduces growth strains (Kakiuchi et al., 2023) and, through molecular dynamics (MD) simulations, enhances clustering, vacancy diffusion, and precipitate formation (Hulse and Race, 2021; Christensen et al., 2018).

We proceed to quantitatively compare our simulation outcomes with experimental findings. The simulations commence beyond a dose threshold where quasi-steady state is achieved, and the nucleation of $\langle a \rangle$ dislocation loops and vacancy clusters concludes. Initially, at 0.5 dpa, the immobile defect densities were $4 \times 10^{21} \text{ L/m}^3$ for each $\langle a \rangle$ dislocation loop, with an average size of 10.9 nm. Correspondingly, for the initial $\langle c \rangle$ bi-pyramids, the immobile defect densities were $4 \times 10^{21} \text{ L/m}^3$, with an average size of 2.4 nm. Table 2 presents a summary of available experimental data on $\langle a \rangle$ and $\langle c \rangle$ dislocation loop size and density, alongside the simulated results. Small vacancy bi-pyramids with a mean size of less than 6 nm are filtered out when determining the mean size from the simulation. The interstitial $\langle a \rangle$ loops' mean size and density are found to closely align with the experimental values. Additionally, the interstitial $\langle a \rangle$ loops exhibit nearly constant mean size, growing slightly from 10.9 nm to 15.1 nm between 0.5 dpa and 20 dpa, consistent with experimental observations showing minimal change in $\langle a \rangle$ loop size with increasing irradiation dose (Kobylyansky et al., 2008; Harte et al., 2017; Allen et al., 2015). The assumption of no nucleation supports the reasonable growth rate of $\langle a \rangle$ loops, as experiments indicating less change in $\langle a \rangle$ loop mean sizes also experience slight nucleation (Harte et al., 2017; Kobylyansky et al., 2008).

Vacancy clusters, which initially had an average size of 2.4 nm, evolve into pure prismatic dislocation loops at 14 nm, as illustrated in Fig. 3(b). This aligns well with atomistic simulations and experiments unable to identify $\langle c \rangle$ dislocation loops smaller than 10 nm (Christiaen et al., 2019; Onimus et al., 2020b; Griffiths, 1988). Recent experiments identifying triangular vacancy platelets (TVPs) as potential precursors to $\langle c \rangle$ dislocation loops corroborate that vacancy loops can nucleate from clusters upon reaching a critical size (Liu et al., 2020). Considering that the maximum size of TVPs was 11 nm, with no vacancy loops found below a diameter of 13 nm, the critical size observed in Fig. 3(b) is reasonable. Moreover, the vacancy $\langle c \rangle$ loops' mean size and density closely match the experimental values, and the $\langle c \rangle$ loops' growth rate is consistent with experiments involving Zr Ion irradiated Zr-4 (Tournadre et al., 2012). For neutron irradiation experiments to doses exceeding 15 dpa, average $\langle c \rangle$ loop sizes were typically 100 nm or larger, with a variance of approximately ± 50 nm. Additionally, $\langle c \rangle$ loop density ranged from 10^{20} to 10^{21} L/m^3 . Consequently, we slightly overestimate the density while slightly underestimating the size of $\langle c \rangle$ dislocation loops. It is important to note that our simulation is calibrated to experimental data by Carpenter et al. (1981) for annealed single crystal Zr growth, while the compared experimental data pertains to polycrystal Zr-2 and Zr-4 (Harte et al., 2017; Tournadre et al., 2012). It is also noteworthy that the coalescence of dislocation loops is not considered.

4.2. Single crystal size effect

In this investigation, we study the influence of equiaxed single crystal size on dislocation loop growth and the resulting growth strains. Fig. 5(a) illustrates growth strains for single crystals of varying sizes (0.2, 1, and 5 μm), using parameters consistent with those outlined in Table 1. Additionally, Fig. 5(b) presents the mean $\langle c \rangle$ defect cluster size, considering bi-pyramids without filtering. It is noted that the mean $\langle a \rangle$ loop sizes across all simulation volumes remain relatively constant, with a slightly higher growth rate observed for larger volumes. Particularly, below 1 μm , notably at 200 nm, growth along both the $\langle a \rangle$ and $\langle c \rangle$ axes is substantially suppressed. This suppression is pronounced by a delayed breakaway along the $\langle c \rangle$ axis and a concurrent decrease in growth strain along the $\langle a \rangle$ axis per dpa. Crystal boundaries act as neutral sinks for mobile defect populations, promoting recombination at these sites and diminishing the number of growing bi-pyramids as well as the influx of vacancies to interior defects. Consequently, a marginal change in the $\langle c \rangle$ cluster size is observed until 6 dpa, after which dislocation loops begin to form within the simulation volume. This coincides with a slight increase in mean cluster size and a gradual breakaway growth along the $\langle a \rangle$ axis and $\langle c \rangle$ axis after 6 dpa.

Above a size of 1 μm , notably at 5 μm , the breakaway growth dose is slightly delayed along both axes, while the growth strain rate is more pronounced beyond the breakaway dose. The transition from stage II to stage III occurs more rapidly, accompanied by a higher growth rate during this transition phase. Although the dose rate and duration of transition to breakaway are affected, the growth strains remain relatively unaffected by increasing crystal size, up to 9 dpa. Nonetheless, at 9 dpa, the larger crystal experiences slightly higher $\langle a \rangle$ axis and $\langle c \rangle$ axis strain. While the growth strain might not exhibit significant alterations due to larger crystals up to 9 dpa, it is crucial to note that the spatial distribution of loop size is contingent on the grain size (Dunn et al., 2016), and the breakaway dose is influenced by the grain size.

To describe the differing spatial distribution of immobile defects from different crystal sizes, kernel density estimations (KDE's) of the vacancy clusters and $\langle c \rangle$ dislocation loops sizes throughout the volume are shown in Fig. 6, along with insets of the vacancy defect distributions at 9 dpa, looking at the basal pole [0001]. The KDE's show an estimation of the probability density function, $f(d)$, where the probability for finding a defect cluster between a size of s_0 and s_1 in the volume is $P(s_0 < s < s_1) = \int_{s_0}^{s_1} f(s)ds$. One of the main features these distributions show is multiple modes of the size distribution, where some experiments show a single mode (Northwood et al., 1979; Gilbert et al., 1979). For the smaller crystal of 200 nm, as shown by Fig. 6(a), the surface-to-volume ratio of boundaries is large as compared to the 1 μm and 5 μm cases, and the formation of $\langle c \rangle$ dislocation loops inside the crystal is inhibited. By 9 dpa, only vacancy clusters around the center of the grain can grow enough to transform into $\langle c \rangle$ dislocation loops. In the center of the volume there is still no growth of vacancy clusters, as the mobile populations behave nearly isotropic, but now near the boundaries there is also no cluster growth, as the boundaries promote recombination of mobile defects. The KDE shows a small fraction of dislocation loops compared with bi-pyramids at 9 dpa, and the dislocation loops that do form grow to between 12–17 nm. No dislocation loops have formed at 5 dpa.

Now, for a simulation size of 1 μm , as shown by Fig. 6(b), we find that at 5 dpa, already a large fraction of the vacancy clusters have become loops. The boundary effect is not strong enough to completely inhibit $\langle c \rangle$ loop formation before 5 dpa, so some begin to form earlier on, and the breakaway dose is 3 dpa, as compared to 6 dpa for the 200 nm size. The KDE is broad and extends between

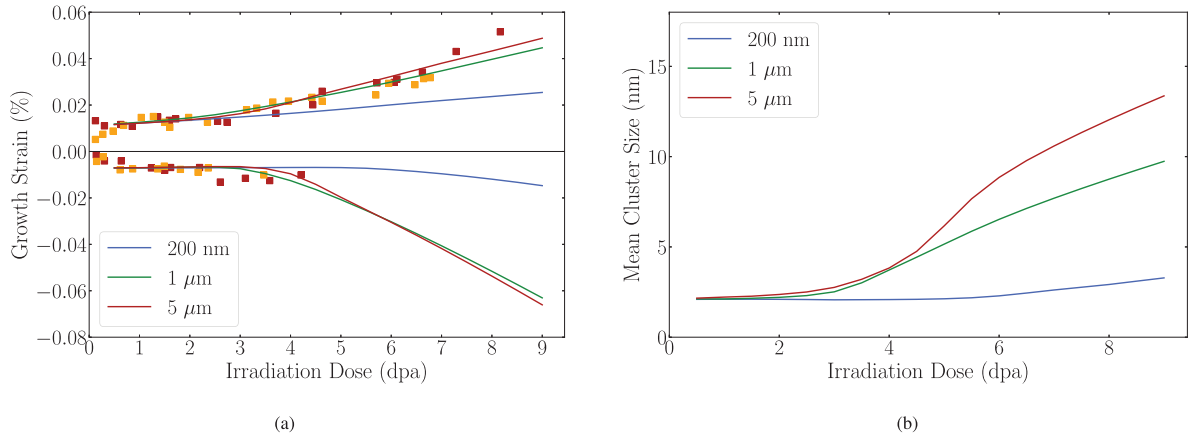


Fig. 5. (a) Simulated growth strains for increasing grain size with the same initial defect density and size. Experimental growth strains from Carpenter et al. (1981) are plotted. (b) Vacancy $\langle c \rangle$ cluster mean size as a function of dpa for a crystal size of 0.2, 1, and 5 μm , including bi-pyramids and loops.

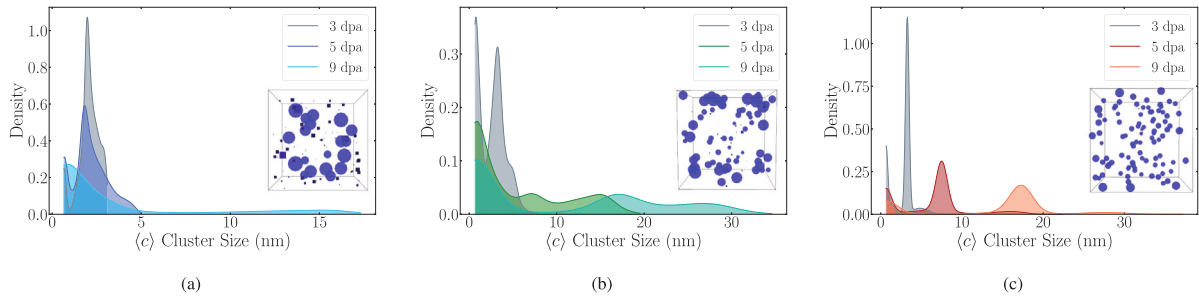


Fig. 6. Immobile cluster gaussian kernel density estimations (KDE), representing the size distribution of $\langle c \rangle$ defects within the simulation volume for different crystal sizes. The spatial distribution is shown by insets at 9 dpa for $\langle c \rangle$ bi-pyramids and loops, viewing parallel to the basal pole [0001]. (a) Size = 200 nm. Inset scaled $\times 2$. (b) Size = 1 μm . Inset scaled $\times 4$. (c) Size = 5 μm . Inset scaled $\times 16$.

a 10–30 nm loop size at 9 dpa, which shows more significant spatial variations in the size distribution of loops, highlighted by the inset. The loop size decreases towards the center of the simulation volume, however, we find a higher density of loops near the boundaries, compared with the 200 nm case.

Increasing the crystal size further to 5 μm , as shown by Fig. 6(c), the spatial distribution of mobile defect concentrations becomes more uniform, as more defect clusters are able to grow, but compete with each other and grain boundaries for vacancies. At 5 dpa, many of the initial bi-pyramids are able to grow to 7 nm, but a very small fraction of vacancy clusters have become loops to around 17 nm in size. With boundary effects essentially negligible, a larger fraction of vacancy clusters grow together slowly, and the competition for vacancies delays the breakaway dose. Upon reaching a critical dose, many vacancy clusters collapse over a short range of dpa, causing the high growth rate, corresponding to the large change in immobile $\langle c \rangle$ defects mean size between 4–6 dpa in Fig. 5(b). At 9 dpa, we find most of the loops are within a size range of 14–21 nm, and a small fraction have grown to >25 nm. The sharper peaks for the KDE for larger grains demonstrate the more uniform spatial variation of bi-pyramids and loops as compared with the 1 μm case.

For a given initial loop density and size determined by the temperature, along with material parameters specified in Table 1, indicating the matrix chemistry, a critical crystal size emerges. This critical size demarcates thresholds below and above which breakaway growth, or stage III, can be either delayed or accelerated. In larger grains surpassing this critical size, the delay arises from the competition among a substantial fraction of growing clusters for vacancies. However, this delay does not mitigate growth strains post-breakaway dose. During the transition from stage II to stage III in larger crystals, the growth rate notably escalates. Conversely, in small nano-crystals beneath the critical size, a pronounced boundary layer effect restricts the flux of mobile defects to immobile vacancy clusters, consequently postponing the breakaway dose. Here, growth strains are significantly diminished due to the inhibition of $\langle c \rangle$ loop formation from vacancy clusters.

In the intermediate regime, around the critical crystal size, a smaller fraction of bi-pyramids exhibit accelerated growth since there is less competition for vacancies and the boundary effect is relatively weak. Consequently, in regions between the center of the crystal and boundaries parallel to the basal pole, certain immobile clusters collapse early, triggering accelerated breakaway compared to both larger and smaller volumes. The transition from stage II to stage III is also more gradual compared to larger volumes. While limited experimental data exists regarding the distribution of loops in irradiated materials, particularly concerning

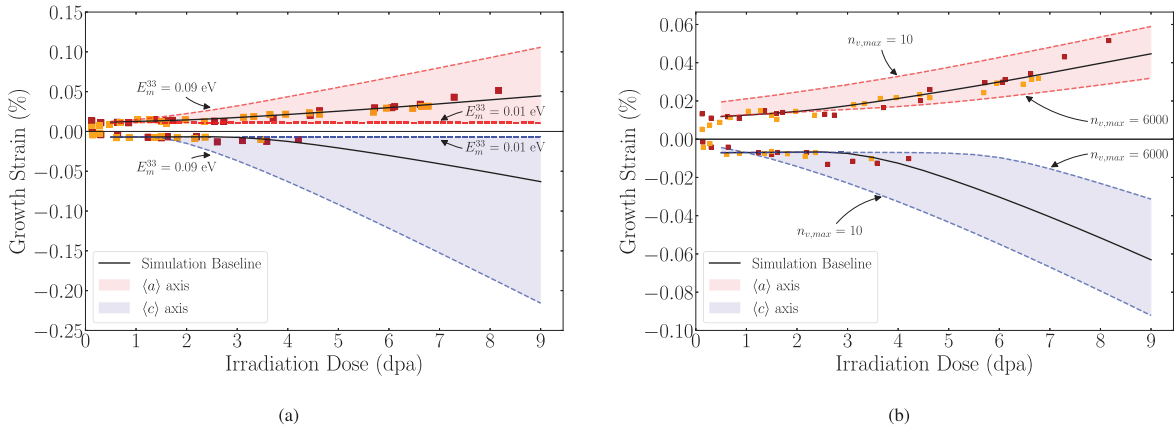


Fig. 7. (a) Effects of di- and tri-interstitial migration energy (DAD) in and out of the basal plane on growth strains. (b) Growth strain as a function of dose for different bi-pyramid-to-loop critical vacancy cluster size.

grain size dependency, the presented results indicate a general increase in the average $\langle c \rangle$ loop diameter with higher grain size at high dose, as illustrated in Fig. 5(b). This observation aligns with experimental TEM findings for irradiated nanocrystalline Mo (Cheng et al., 2016), which also indicate an increase in loop diameter with grain size. Additionally, the results underscore that nanocrystals (with an average crystal size below approximately 500 nm) exhibit significantly reduced irradiation growth strains for single crystals, as evidenced by the 200 nm case.

4.3. The influence of diffusion anisotropy & the bi-pyramid-to-loop transition

The Diffusion Anisotropy Difference (DAD), quantified by the parameter $p^m = (D_c^m/D_a^m)^{1/6}$, is essential for describing the preferential diffusion of mobile defects within the basal plane (Woo, 1988). This parameter has been shown to be necessary to describe the breakaway growth phenomenon observed in stage III (Zhang et al., 2023; Li et al., 2020; Barashev et al., 2015; Christien and Barbu, 2009). As depicted in Fig. 7(a), when $p^m = 1$ (i.e., $E_m^{33} = 0.01$), the migration of di- and tri-interstitials becomes isotropic, effectively halting breakaway growth. Conversely, reducing the anisotropy factor p^m (i.e., $E_m^{33} = 0.09$), hinders the diffusion of these defects out of the basal plane, thereby increasing $\langle a \rangle$ strains during both steady-state and breakaway stages. This adjustment leads to an accelerated onset of breakaway growth and larger growth strains at higher irradiation doses.

The transition from vacancy bi-pyramids to dislocation loops is equally crucial in understanding breakaway growth phenomena. As illustrated in Fig. 7(b), in the absence of bi-pyramids, with vacancy loops forming almost immediately at a size of 10 vacancies, a nearly constant growth strain rate is observed. This scenario fails to capture the quasi-steady growth observed during stage II from 1–3 dpa along the $\langle c \rangle$ axis. The growth strains are much larger at each dpa, as sessile dislocation loops are more effective in absorbing mobile defects than immobile clusters. Furthermore, the $\langle a \rangle$ axis initially experiences a larger growth strain, as no bi-pyramids are compensating for the $\langle a \rangle$ axis expansion. Extending the bi-pyramid to loop transition over a broader range of dpa leads to breakaway growth occurring at higher doses. The experimental data from Carpenter et al. (1981) was only taken up to 4 dpa for the $\langle c \rangle$ axis single crystals, and with the current interpolated cluster regime between 400–1500 vacancies, and initial density and sizes of $4 \times 10^{21} \text{ L/m}^3$ and 2.4 nm, respectively, the breakaway along the $\langle c \rangle$ axis of the crystal occurs between 3–4 dpa. However, extending this regime to include a range of 400–6000 vacancies, delays breakaway to 6 dpa along the $\langle c \rangle$ axis, and reduces $\langle a \rangle$ axis growth strains at each dpa. It is important to note that the critical vacancy cluster size at the transition point may vary, as observed in atomistic simulations for hcp Zr (Christien et al., 2019), where different methodologies yielded diverse transition ranges. Furthermore, some alloying elements, such as Niobium, can delay the breakaway dose in the fashion observed here.

4.4. Results for polycrystals and the influence of texture

As a result of fabrication of Zr cladding in PWR fuel rods and pressure tubes in CANDU reactors, crystallographic axes acquire preferred orientations, referred to as *texture*. In this section, we study the influence of the specific texture and grain size on predicted irradiation growth. The texture is characterized by f-parameters, or Kearns parameters that are commonly designated as radial (f_r), transverse (f_t), and axial (f_a), and can be calculated from X-ray diffraction and EBSD data. Poles of crystallographic orientations are distributed in terms of volume fraction about those directions in the cladding tube. The f-parameters of a typical PWR cladding for the [0001] basal pole are: $f_t = 0.317$, $f_r = 0.636$, and $f_a = 0.047$, while for CANDU pressure tubes they are: $f_t = 0.61$, $f_r = 0.33$, and $f_a = 0.06$ (Adamson et al., 2019). It is noted that grain boundaries are conceptualized as perfect sinks for mobile defects in the sense that their formation energy g is assumed to be zero and independent of the grain boundary type. Hence grain boundaries will effectively absorb, vacancies, and interstitials without any impediment. To account for specific boundary effects (e.g. grain boundary types, defect accumulation, etc.) the defect formation energy must be modeled accordingly. For simplicity, our current

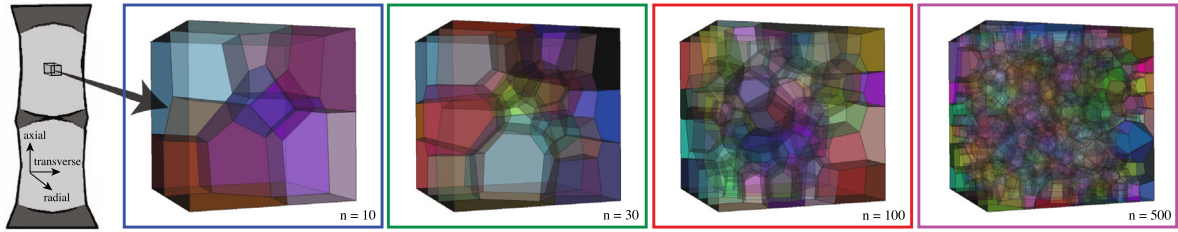


Fig. 8. Adapted schematic diagram of the cladding (Alam et al., 2011), with labeled axial, transverse, and radial directions. Polycrystalline simulation volumes are shown, with 1 μm box side length, and varying the number of grains (10, 30, 100, 500), resulting in varying average grain size (464 nm, 322 nm, 215 nm, 126 nm), respectively. For each case, simulation volumes were prescribed PWR f-parameters, where $f_r = 0.63$, $f_t = 0.32$, $f_a = 0.05$, corresponding to the fraction of basal poles [0001] oriented along each direction of the cladding.

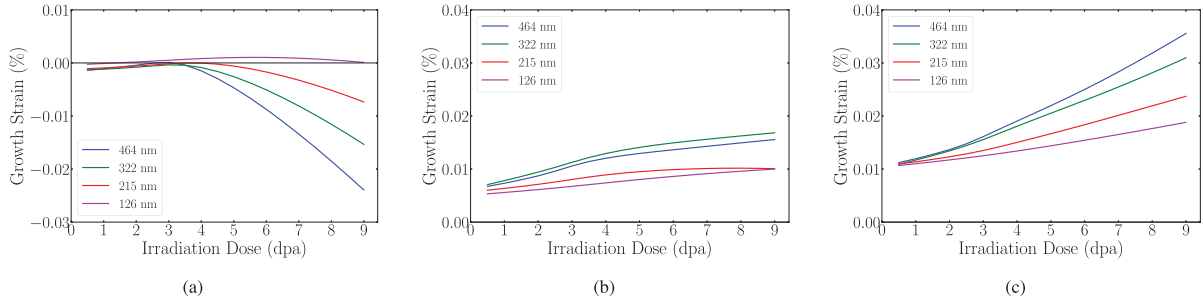


Fig. 9. Growth strains varying grain size with PWR f-parameters $f_r = 0.63$, $f_t = 0.32$, $f_a = 0.05$. Each curve is an average of three simulations. (a) Growth strains along the radial direction. (b) Growth strains along the transverse direction. (c) Growth strains along the axial direction. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

model does not consider these additional effects in Eq. (32). The PWR texture is approximated by selecting a fraction of grains with basal poles in each of the three global coordinate axes, shown by Fig. 8. A grain with the basal pole in the radial direction will have its [0001] crystallographic axis parallel to the radial axis of the cladding tube, $[2\bar{1}10]$ axis parallel to the transverse axis, and $[01\bar{1}0]$ axis parallel to the axial axis. This follows similarly for grains with their basal pole parallel to the transverse or axial axes. The grain size is normally distributed.

Furthermore, LWR cladding are equiaxed, with grain size in the range between 2–9 μm (Adamson et al., 2019; Kobylansky et al., 2008). Here we study grain sizes approaching 500 nanometers. Since PWR and CANDU textures are related by a rotation, only one of the orientations is studied. The PWR texture for a grain of the cladding ($f_t = 0.317$, $f_r = 0.636$, and $f_a = 0.047$) was randomized and the order of the rotation matrices for each grain was also randomized during each simulation. This insures that computer-generated grains satisfy the Kearns parameters. Three simulations were performed for each grain size with PWR texture, changing the orientations of the grains with a different seed, but maintaining PWR f-parameters. The growth strains were then averaged between the three simulations for each grain size. Each simulation has a box of 1 μm side length, thus, increasing number of grains has the effect of decreasing the average grain size.

Fig. 9 shows the irradiation growth strains as function of dose for polycrystals of average grain sizes of 464 nm (blue), 322 nm (green), and 215 nm (red), 126 nm (purple). In Fig. 9(a) growth strains are shown along the radial direction, in Fig. 9(b) along the transverse direction, and in Fig. 9(c) along the axial direction. Fig. 10 shows the vacancy clusters and dislocation loops spatial distribution at 9 dpa for the simulations with average grain sizes of 464 nm (blue) and 215 nm (red). In Fig. 10(a) the discrete representation is shown looking at the radial direction, in Fig. 10(b) looking at the transverse direction, and in Fig. 10(c) looking at the axial direction. Decreasing the average grain size yet maintaining PWR texture parameters shows a significant reduction in growth strain along all principal axes, compared to the single crystal results. Below 215 nm, only the radial directions sees a further improvement in growth resistance. The introduction of more grain boundaries improves irradiation damage resistance as abundant grain boundary surfaces behave like perfect sinks for mobile defects. The radial direction shows the most significant improvement as fewer bi-pyramids collapse into loops up to 9 dpa. The introduction of texture also improves irradiation damage resistance, with the most significant improvements along the radial and transverse directions. Since very few basal poles are oriented along the axial direction of the cladding, growth resistance is not improved much in this direction. Both the introduction of grain boundaries and texture can help in reducing irradiation growth along directions with high Kearns parameters. Randomly oriented grains in the transverse direction ($f_t = 0.32$) show the most resistance to growth, consistent with experimental observations (Kobylansky et al., 2008; Adamson et al., 2019).

Experimental studies have established a correlation between growth strain in irradiated polycrystalline materials and a growth anisotropy factor, G_d , where $G_d \propto 1 - 3f_d$, with f_d representing the Kearns parameter along a given direction (Adamson et al.,

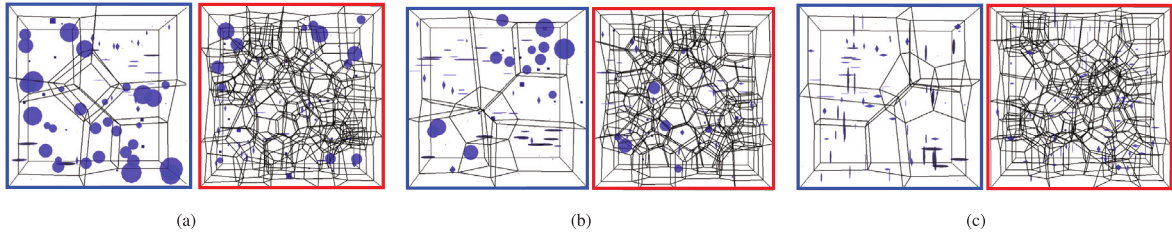


Fig. 10. Vacancy clusters and loops at 9 dpa for simulations with average grain size of 464 nm (blue border) and 215 nm (red border) given PWR f -parameters $f_r = 0.63$, $f_t = 0.32$, $f_a = 0.05$. (a) Viewing along the radial direction. Scaled $\times 4$ (b) Viewing along the transverse direction. Scaled $\times 4$ (c) Viewing along the axial direction. Scaled $\times 4$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

2019). Our simulations reveal that the most significant resistance to irradiation growth occurs in the transverse direction of the cladding, characterized by a Kearns parameter of $f_t = 0.32$ (see Fig. 9(b)). Conversely, for directions with smaller f values, such as the axial direction ($f_a = 0.05$), G_d tends to be positive and large (see Fig. 9(c)), whereas for larger f values, as in the radial direction ($f_r = 0.63$), G_d tends to be negative and large (see Fig. 9(a)). This validation of the growth anisotropy factor, $G_d \propto 1 - 3f_d$, underscores its utility in describing annealed polycrystalline Zr behavior under irradiation. However, deviations from this factor are observed for very small grain sizes, particularly along the radial direction. At higher doses, the growth rate may shift as some bi-pyramids transform into loops, a phenomenon more prevalent in larger grains.

Nanograin polycrystalline materials have demonstrated promising resistance to irradiation damage (Singh, 1973; Bai et al., 2010; Sun et al., 2013; Beyerlein et al., 2013; Jin et al., 2018; Su et al., 2023), with efficacy depending on grain size and the specific radiation-induced phenomena under investigation. Notably, when the mean free path of point defects created during a cascade is comparable to the average grain size, significant damage alleviation can occur (Zhang et al., 2018). High-angle grain boundaries have also been observed to absorb dislocation loops in their vicinity (Sun et al., 2013). However, it is important to note that nanocrystals with superfine nanograins may become easily amorphous and less effective in resisting radiation damage (Chang et al., 2013). We have found that the density and growth rate of basal and prismatic loops drastically reduces with decreasing grain size, consistent with experimental studies of irradiated nanocrystalline materials, and other cluster dynamics simulations (Sun et al., 2013; Cheng et al., 2016; Dunn et al., 2016). The results presented in this section underscore the inhibition of irradiation growth rates in smaller grain-size materials compared to single crystals. As average grain size decreases, irradiation growth strain at a specific dose also decreases, highlighting the crucial role of grain boundary surfaces as neutral sinks for mobile defects.

5. Conclusions

We presented here a novel computational approach that connects the mechanics of inelastic deformation to the theory of radiation damage in a way that is computationally tractable. While early efforts in radiation damage theory did not consider at all the spatial dependence of damage evolution in materials, more recent treatments emphasized the solution of unnecessarily large sets of ODEs and were limited to 1-dimensional geometry. This approach, again, also makes it nearly impossible to tackle more realistic mechanical deformation problems. Fortunately, nature has provided us with the principle of *separation of scales*, where one can eliminate fast variables (both in size and in time domain) adiabatically. We utilized this principle twice here. First, in the time domain, we eliminated the fast variables of mobile defect concentrations by assuming that they are always at a state of *quasi-steady-state*, in equilibrium with the slow variables. These are the components of the microstructure that grow slowly, i.e. dislocation loops in the current treatment. Second, we correspondingly divided the interacting defect species populations into two groups; one mobile and the other sessile. The sessile population was described by an average size and density, thus simplifying the way that non-linear reactions are calculated. With these simplifications, we were able to achieve two goals: (1) the ability to cast the diffusion problem, with multiple non-linear reactions, into a weak variational form amenable to FE solutions in arbitrary 3-D geometry, and (2) the direct coupling to mechanics of inelastic deformation with fully explicit representation of complex grain structure, boundary conditions, and loading scenarios.

The theoretical and computational advances made here enabled us to address important questions relating to the influence of grain structure, grain boundary content, nanograin size, and mechanical texture on the irradiation growth of Zr. The coupled CD-plasticity model also provides a tool for investigating other radiation damage phenomena that are essentially mechanical in nature, such as creep and swelling. When applied to the case of irradiation growth in Zr alloys, our CD plasticity model leads to the following conclusions.

1. The simulated growth strains are in agreement with experimental growth data for single crystal Zr from Carpenter (Fig. 3(a)), and all parameters are within reasonable ranges, compared with experimental data and atomic simulations (Table 1). The transformation of a vacancy cluster into a full vacancy loop at a size of 14 nm is in good agreement with experimental observations and atomistic simulations (Fig. 3(b)). The density, size and growth rate of the $\langle c \rangle$ and $\langle a \rangle$ dislocation loops agree well with the TEM of irradiated Zr and its alloys (Table 2).

2. The growth of $\langle c \rangle$ and $\langle a \rangle$ dislocation loops is strongly dependent on spatial location, as observed in TEM micrographs of irradiated Zr. The spatial density of the vacancy and prismatic loops can vary significantly in samples irradiated to high doses (Fig. 4).
3. Growth strains saturate up to 9 dpa when increasing crystal size, however, the growth of dislocation loops varies spatially depending on crystal size (Fig. 6) for a given material and temperature. In smaller crystals, the diffusive boundary flux becomes significant enough to cease or slow the growth of loops, only allowing a few vacancy clusters to grow slowly, and delays the breakaway significantly. Boundaries modeled as unbiased sinks promote the recombination and absorption of mobile defects near them, leaving fewer defects available for immobile species. In very large crystals, a large fraction of the vacancy clusters in the volume are able to grow and compete for vacancies, slowing their individual growth and causing the transformation and thus breakaway to happen at a slightly later dose. The large number of growing clusters will collapse together and a larger growth strain rate and thus steeper breakaway are observed.
4. The diffusion anisotropy difference (DAD) of mobile interstitial species is necessary to describe the growth behavior of the $\langle a \rangle$ axis expansion and the $\langle c \rangle$ axis contraction in irradiated Zr (Fig. 7(a)). Under isotropic conditions, mobile species do not see an effective difference between interstitial and vacancy-type dislocation loops, resulting in no absorption of mobile species and no growth.
5. The bipyramid-to-dislocation-loop transition is necessary for breakaway growth along the $\langle c \rangle$ axis. Furthermore, steady-state growth along the $\langle a \rangle$ and $\langle c \rangle$ axes cannot be captured without the transition function (Fig. 7(b)). Bipyramids have isotropic eigenstrains, and it is not until their collapse into $\langle c \rangle$ dislocation loops that large growth strain rates can be observed along both crystal axes.
6. It has been verified that the growth strains of annealed polycrystalline Zr can be approximated using Kearns parameters with the relationship $G_d \propto 1 - 3f_d$. The parameters of PWR Kearns, $f_r = 0.63$, $f_i = 0.32$, $f_a = 0.05$, show improved resistance to irradiation along the principal directions compared to single crystals with the same simulation volume. Reducing grain size to nanograins showed more resistance to irradiation-induced growth, particularly along the direction with the highest volume fraction of the basal poles [0001] (Figs. 9, 10).

CRedit authorship contribution statement

Matthew Maron: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yang Li:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Inam Lalani:** Writing – original draft, Investigation. **Kristopher Baker:** Writing – review & editing, Methodology, Conceptualization. **Benjamin Ramirez Flores:** Writing – review & editing, Methodology, Conceptualization. **Thomas Black:** Writing – review & editing, Methodology, Conceptualization. **James Hollenbeck:** Writing – review & editing, Methodology, Conceptualization. **Nasr Ghoniem:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Investigation, Conceptualization. **Giacomo Po:** Writing – review & editing, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no conflict of interests.

Data availability

Data will be made available on request.

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Appendix A. Discussion of the bias factors

In the framework of rate theory or cluster dynamics, the defect absorption rate by sinks is quantified by the so-called absorption efficiencies. They are obtained by solving the diffusion problem in the vicinity of a sink through an effective medium at a stationary state. Analytical expressions of absorption efficiencies can be obtained only with simple geometries and simplified description of point defect properties (Woo and Goesele, 1983; Woo, 1988). The full complexity of this problem entails dedicated numerical simulations, such as molecular dynamics (Wolfer, 2007) and kinetic Monte-Carlo (Da Fonseca et al., 2023) simulations.

It is widely accepted that in describing the absorption efficiency of point defects, namely interstitials and vacancies, to dislocations, the elastic interaction between defects and dislocations, the diffusional anisotropy of defects, and external stresses

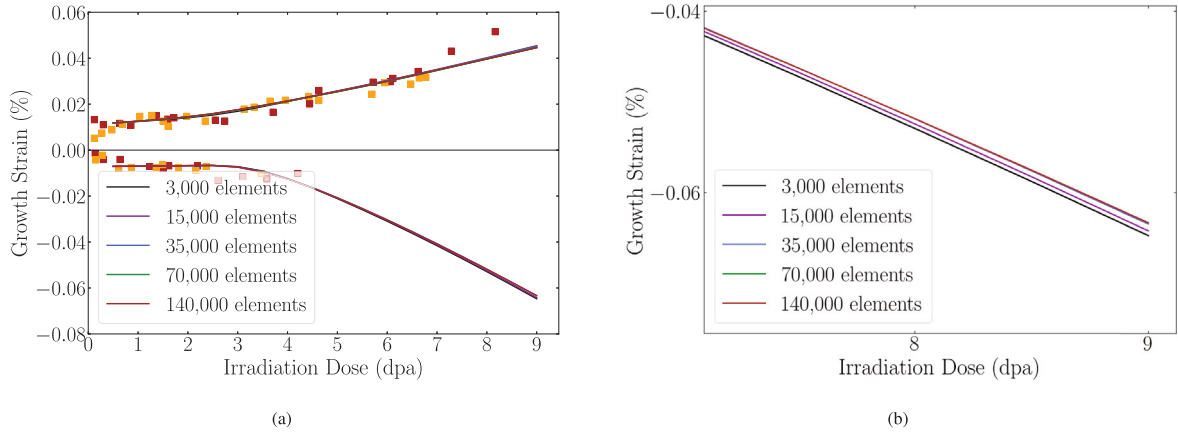


Fig. B.11. Mesh size study from 3000 to 140,000 elements to verify convergence of simulated growth strains. (a) Full growth strain plots of <a> and <c> axis strains. (b) Zoomed depiction of <c> axis growth strains verifying convergence at a mesh size of 70,000 elements in a 1 μm size simulation domain.

are the most important factors affecting the absorption rate. First, the elastic interaction of point defects with dislocations modifies the migration of the point defects towards the sinks and thus their reaction rates. The vacancies and interstitials have different interaction energies with dislocations, which ultimately gives rise to the different absorption efficiencies. This is the origin of the conventional dislocation bias. Second, in the presence of the external applied stress, it is found that the absorption efficiency depends on the relative orientation of dislocations with respect to the external stress. The stress and orientation dependence of the absorption rates at dislocations give rise to a stress-induced preferred absorption, known as SIPA, and often used to explain irradiation creep. An applied stress can thus modify the dislocation-point defect interactions. Considering a defect as a spherical inhomogeneity, the long-range strain field of a dislocation can detect the perturbation of the applied stress field near the point defect, and then gives rise to an inhomogeneity interaction. This interaction enables point defects to distinguish which are the more favorable dislocations to migrate towards. By assuming that the dipole tensor is the same at the stable and saddle points, [Heald and Speight \(1975\)](#) have shown that this effect, usually denoted as SIPA-I, can be characterized by a factor $Z^I(\sigma)$ as

$$Z^I(\sigma) = Z^0 \left(1 + \frac{\Delta Z^I(\sigma)}{Z^0} \right) \quad (\text{A.1})$$

where Z^0 is the absorption efficiency without applied stress. $\Delta Z^I(\sigma)$ is the deviation due to the inhomogeneity interaction, depending on the defect property, the stress, and the dislocation direction. On the other hand, an applied stress can also influence the jump barrier of point defects as a function of the jump direction. This is the so-called elastodiffusion (SIPA-AD), and has been thoroughly studied by [Woo and Goesele \(1983\)](#) and [Borodin and Ryazanov \(1994\)](#), i.e.,

$$Z^{AD}(\sigma) = Z^0 \left(1 + \frac{\Delta Z^{AD}(\sigma)}{Z^0} \right) \quad (\text{A.2})$$

where $\Delta Z^{AD}(\sigma)$ describes the effects of the saddle point anisotropy due to the stress. Contrary to the SIPA-I model, defects have different properties at stable and saddle positions in this model.

In addition, the crystallographic anisotropy of non-cubic metals makes the diffusion rates of defects dependent on the crystallographic directions even in the absence of the external applied stress as well. A difference between the diffusional anisotropy of the vacancy and interstitial would hence result in a difference in the absorption rates at dislocations, which is known as diffusional anisotropy difference (DAD). It should be pointed out that this DAD in non-cubic crystal is a zero-order effect associated with the intrinsic diffusional anisotropy, and is distinguished from the aforementioned elastodiffusion. According to [Woo \(1988\)](#), DAD effect of dislocations can be simply expressed as

$$Z^{DAD,0} = Z^0 p^{-2} (\cos^2 \theta + p^6 \sin^2 \theta)^{\frac{1}{2}} \quad (\text{A.3})$$

where $p = (D_c/D_a)^{1/6}$ is the diffusional anisotropy factor of the defect and θ is the angle between the symmetry axis (e.g., c -axis for HCP crystals) and the dislocation line direction. The DAD concept has been successfully adopted to describing the irradiation growth in zirconium. For prismatic and basal loops, it reduces to

$$Z_{a\mathcal{L}}^{DAD,0} = \frac{p + p^{-2}}{2} \quad (\text{A.4})$$

$$Z_{c\mathcal{L}}^{DAD,0} = p \quad (\text{A.5})$$

In summary, a thorough description of the irradiation deformation requires an accurate evaluation of the absorption efficiency of sinks considering all above effects as

$$Z^{tot} = Z^{DAD,0} \left(1 + \frac{\Delta Z^I(\sigma)}{Z^{DAD,0}} + \frac{\Delta Z^{AD}(\sigma)}{Z^{DAD,0}} \right) \quad (\text{A.6})$$

Appendix B. Convergence study

As a first test to the SR-CD framework we perform a convergence study where we meticulously examined the impact of varying mesh sizes on the accuracy of our computational model. The analysis spanned a range from 3000 to 140,000 elements, for a 1 μm size simulation domain, focusing on the influence to irradiation dose responses as shown in Fig. B.11. The results of this analysis demonstrated that variations in mesh size did not largely affect the calculated growth strains, indicating that the mesh size of 70,000 elements was sufficiently fine to capture the essential physics of the problem without introducing numerical artifacts or significant computational overhead. Our findings demonstrated a significant stabilization in the growth strain measurements which indicate that a mesh size of 70,000 elements was sufficiently accurate for our purposes.

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